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# Catching a Moving Subspace: Low-Rank Bandits Beyond Stationarity

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## Abstract

Many real bandit deployments (recommendation, clinical dosing, ad targeting) share two structural facts handled only in isolation by prior work: rewards live on a low-dimensional latent subspace, and that subspace drifts. Stationary low-rank bandits exploit the rank but break under any subspace change; non-stationary linear bandits adapt to drift but pay the ambient rate  $\tilde{O}(d\sqrt{T})$ . We study *piecewise-stationary low-rank* linear contextual bandits with scalar feedback:  $\theta_t = B_k^* w_t$  for a rank- $r$  factor  $B_k^* \in \mathbb{R}^{d \times r}$  that is constant within each of  $K$  unknown segments and may shift at the boundaries. Our results are tight along three axes. **(i) Identification boundary.** With single-play scalar rewards, the moving subspace is recoverable through quadratic functionals of rewards if and only if three probe-side conditions hold (known noise variance, bounded state-noise coupling, full-dimensional probe support). Each is individually necessary in the unrestricted-second-moment problem, and the three are jointly sufficient: a characterization of the boundary of the solvable region, not just a sufficient interior point. **(ii) Algorithm and dynamic regret.** SPSC (Single-Play Subspace-Calibrated Optimism) interleaves isotropic probes with windowed projected ridge-UCB exploitation inside the learned  $r$ -dimensional subspace; a CUSUM-style adaptive variant discovers segment boundaries online. The costed dynamic regret is  $\tilde{O}(r\sqrt{T}) + \tilde{O}(T^{2/3}) + O(W V_{\text{in}})$ , replacing the ambient  $d\sqrt{T}$  rate with the intrinsic rank. **(iii) Empirics.** On eleven benchmarks spanning synthetic, UCI/MovieLens, semi-synthetic clinical, and ZOZOTOWN production-log data, SPSC outperforms non-stationary and low-rank baselines whenever  $d - r \gtrsim T^{1/6}$ , matching the analytical crossover. To our knowledge, this is the first work to characterize the identification boundary and attain the intrinsic-rank dynamic-regret rate in this setting.

## 1 Introduction

High-dimensional sequential decisions — personalized recommendation, clinical dosing, ad allocation — share two structural facts that the linear-bandit literature has so far treated only in isolation. The reward parameter  $\theta_t \in \mathbb{R}^d$  at round  $t$  typically admits a *low-rank* factorization  $\theta_t = B_k^* w_t$  with  $r \ll d$ : user preferences, clinical responses, and ad-creative effects live on a thin slice of the ambient space. That thin slice is *not static*: tastes drift, cohorts change, treatment regimes get revised, and the subspace  $\text{range}(B_k^*)$  itself moves across unknown change points  $1 = \tau_0 < \tau_1 < \dots < \tau_K = T+1$  that delimit the segments  $\mathcal{I}_k = [\tau_{k-1}, \tau_k)$ . Production deployments make both facts visible: Spotify’s homepage contextual bandit explicitly targets daily shifts in user intent [Feijer et al., 2025]; the IWPC

warfarin cohort [Consortium, 2009] has been re-analyzed to expose three latent patient subgroups whose dose-response coefficients flip sign across boundaries [Liu et al., 2025]; the public Open Bandit logs [Saito et al., 2021] ship scalar feedback collected across multiple ad campaigns at ZOZOTOWN. What is missing is a combined treatment in linear contextual bandits: rewards living on a low-rank subspace whose identity shifts at unknown change points, observed only through scalar feedback  $y_t = x_t^\top \theta_t + \varepsilon_t$ .

**The gap, and why it is non-trivial.** Stationary low-rank bandits [Jun et al., 2019, Jedra et al., 2024, Lu et al., 2021, Jang et al., 2024] deliver the  $\tilde{O}(r\sqrt{T})$  rate but assume a fixed  $B^*$ ; any non-trivial subspace change voids the analysis ( $\tilde{O}$  suppresses polylog factors in  $T, d, r, 1/\delta$ ). Non-stationary linear bandits [Russac et al., 2019, Cheung et al., 2019] absorb drift through sliding windows or discounting but operate in  $\mathbb{R}^d$  and pay  $\tilde{O}(d\sqrt{T})$  within each segment, ignoring intrinsic rank entirely. Naive combinations fail in instructive ways: sliding-window ridge in  $\mathbb{R}^d$  never recovers the subspace; a one-shot subspace estimate breaks the moment a change point fires; per-segment re-runs of stationary low-rank algorithms require oracle access to the boundaries. The genuine difficulty is twofold. (a) With scalar rewards, a single response  $y_t$  is a rank-one quadratic measurement of  $\theta_t \theta_t^\top$ , so the subspace is recoverable only through *second-order* probing. (b) Every probe diverts a round away from exploitation, so probe acquisition must itself be priced against the regret it enables. No prior algorithm recovers a changing subspace under scalar feedback, and no prior analysis prices that recovery against the exploitation regret it buys.

**Contributions.** We formulate the *piecewise-stationary low-rank* linear bandit (§2) and resolve it on three axes. (i) **Identification boundary (Theorem 2.2).** Three probe-side conditions (known noise variance, bounded state-noise coupling, full-dimensional probe support) are individually necessary in the unrestricted-second-moment problem and jointly sufficient for recovering  $\text{range}(B_k^*)$  from quadratic functionals of scalar rewards. The necessity half — three matching impossibility results (Propositions B.12–B.14) — is what distinguishes this from a generic identifiability statement: it characterizes the boundary of the solvable region, not just a sufficient interior point. (ii) **Algorithm and costed dynamic regret (Theorem 4.1).** SPSC (Alg. 1) interleaves isotropic probes with windowed projected ridge-UCB exploitation inside the learned  $r$ -dimensional subspace via a quadratic-measurement identity; a **SPSC-Adaptive** variant (Alg. 2) uses a CUSUM-style detector to discover segment changes online. The costed dynamic regret is  $\text{DynReg}_T^{(c)} \leq \tilde{O}(r\sqrt{T}) + \tilde{O}(T^{2/3}) + O(W V_{\text{in}})$ , replacing the ambient  $d\sqrt{T}$  rate with the intrinsic rank  $r$ , where  $W$  is the exploitation window and  $V_{\text{in}}$  the within-segment path variation. (iii) **Eleven benchmarks against eleven baselines.** A synthetic phase-transition grid, five UCI environments plus MovieLens, two semi-synthetic clinical datasets, the Russac small- $d$  benchmark, and Open Bandit production logs, against LinUCB, D-LinUCB, SW-LinUCB, Restart-LinUCB, LowOFUL, VOFUL, LowRank-Reward, LinTS, SW-LinTS, and adapted stationary low-rank methods BOSS and Jedra. SPSC delivers regret reductions whenever  $d - r \gtrsim T^{1/6}$ , matching the analytical crossover. A direct necessity stress test (App. G.3, part C) restricts probe coverage to a proper subspace and reproduces the  $\Omega(T)$  blow-up predicted by Proposition B.14, confirming the third identifiability condition is not a proof artifact.

To our knowledge, this is the first work to characterize the identification boundary and to attain the intrinsic-rank dynamic-regret rate in the piecewise-stationary low-rank setting under scalar feedback.

**Related work.** Stationary low-rank bandits [Jun et al., 2019, Kang et al., 2022, Jedra et al., 2024, Jang et al., 2024, Stojanovic et al., 2023, Duong et al., 2024, Lu et al., 2021] achieve intrinsic-rank rates but assume a fixed  $B^*$ ; nonstationary bandits [Russac et al., 2019, Cheung et al., 2019, Abbasi-Yadkori et al., 2023, Hou et al., 2024, Kim et al., 2022] bound dynamic regret in the ambient dimension and ignore low-rank structure; cost-aware observation [Seldin et al., 2014, Tucker et al., 2023, Elumar et al., 2024] prices information acquisition for fixed parameters, not for identifying a changing subspace. SPSC is the first algorithm we are aware of that (i) attains regret scaling with intrinsic rank  $r$ , (ii) adapts to

changing subspaces without oracle boundaries, and (iii) prices probe acquisition explicitly. Appendix A gives the full positioning (Tables 5–7).

## 2 Setting and probe-based identification

**Piecewise low-rank model.** At every round  $t$  the reward parameter admits a factorization  $\theta_t = B_k^* w_t$ , where  $B_k^* \in \mathbb{R}^{d \times r}$  has orthonormal columns and is piecewise constant over  $K \geq 1$  segments  $\mathcal{I}_k$  with unknown change points  $1 = \tau_0 < \tau_1 < \dots < \tau_K = T+1$ , and the latent state  $w_t \in \mathbb{R}^r$  follows a stable linear dynamical system inside each segment

$$w_t = A_k w_{t-1} + \eta_{t-1}, \quad t \in \mathcal{I}_k,$$

with unknown  $A_k$  satisfying  $\rho(A_k) \leq 1 - \alpha_0$  for some  $\alpha_0 > 0$  (used only to ensure  $\|w_t\| \leq S_w$  via the stationary covariance bound) and zero-mean innovation  $\eta_{t-1}$  with covariance  $\Sigma_{\eta,k} \succ 0$ . We assume uniform action-boundedness  $\sup_{x \in \mathcal{A}_t} \|x\| \leq R_{\mathcal{A}}$ , a known probe distribution  $Q$  with  $\mathbb{E}[uu^\top] \succeq \rho_Q I_d$  and sub-Gaussian marginals (bounded  $\|u\| \leq L$  is a special case; see Remark 2.1), conditionally  $\sigma_\varepsilon$ -sub-Gaussian noise with unknown variance, bounded state-noise coupling  $\|\mathbb{E}[\varepsilon_t \theta_t]\|_2 \leq \epsilon_\times$ , and full-dimensional probe coverage. Theorem 2.2 below shows these three probe-side conditions are individually necessary and jointly sufficient for identifying the changing subspace from single-play scalar rewards. Throughout,  $\delta \in (0, 1)$  denotes the global confidence parameter (concentration bounds hold with probability  $\geq 1 - \delta$ ) and  $\lambda > 0$  denotes the ridge regularizer used by the windowed estimator of §3.

**Remark 2.1** (Probe distribution: scaled sphere). *The theory uses scaled-sphere probes  $u_t = \sqrt{d} v_t$ ,  $v_t \sim \text{Unif}(\mathbb{S}^{d-1})$ , which satisfy  $\|u_t\|_2 = \sqrt{d}$  exactly (so no truncation event is needed) and  $\mathbb{E}[uu^\top] = I_d$ . In our implementation we draw  $\tilde{u} \sim \mathcal{N}(0, I_d)$  and rescale  $u \leftarrow \sqrt{d} \tilde{u} / \|\tilde{u}\|$ , which is exactly the scaled-sphere distribution. The same argument applies to isotropic Gaussian probes  $u_t \sim \mathcal{N}(0, I_d)$  via a  $\chi_d^2$  truncation at  $L = \sqrt{2d \log(2T/\delta)}$  (Appendix C).*

**Quadratic measurement identity.** Let  $\hat{\sigma}^2$  be the algorithm’s estimate of the noise variance and define the centered probe statistic  $s_t := y_t^2 - \hat{\sigma}^2$  on every probe round  $t \in \mathcal{T}_{\text{probe}}$ . Under the probe-noise assumptions of §2, for  $\delta_\sigma := \hat{\sigma}^2 - \sigma_\varepsilon^2$ ,

$$\mathbb{E}[s_t \mid \mathcal{H}_{t-1}, u_t] = u_t^\top \widetilde{M}_t u_t + 2u_t^\top \mathbb{E}[\varepsilon_t \theta_t \mid \mathcal{H}_{t-1}, u_t] - \delta_\sigma, \quad (1)$$

where  $\widetilde{M}_t := \mathbb{E}[\theta_t \theta_t^\top \mid \mathcal{H}_{t-1}]$  is the predictable second moment, whose range is contained in  $\text{range}(B_k^*)$  for  $t \in \mathcal{I}_k$ ; under the non-degenerate innovation condition  $\Sigma_{\eta,k} \succ 0$  (§2), the probe-time average has range exactly  $\text{range}(B_k^*)$  (Proposition B.9). Under exact probe conditions ( $\delta_\sigma = \epsilon_\times = 0$ ) this collapses to  $\mathbb{E}[s_t \mid \cdot] = u_t^\top \widetilde{M}_t u_t$ : a scalar reward carries a quadratic measurement of  $\widetilde{M}_t$ .

**Lifted subspace estimator.** The probe moment operator  $\mathcal{K} : M \mapsto \mathbb{E}[(u^\top M u) u u^\top]$  admits a closed-form inverse on symmetric matrices for scaled-sphere probes  $u = \sqrt{d} v$ ,  $v \sim \text{Unif}(\mathbb{S}^{d-1})$ :

$$\mathcal{K}(M) = \frac{d}{d+2} (\text{tr}(M) I_d + 2M), \quad \mathcal{K}^{-1}(N) = \frac{d+2}{2d} N - \frac{\text{tr}(N)}{2d} I_d, \quad \|\mathcal{K}^{-1}\|_{\text{op} \rightarrow \text{op}} \leq 1 \quad (2)$$

on the symmetric subspace (Lemma B.4). Define the *lifted probe sample*

$$G_t := \mathcal{K}^{-1}(s_t u_t u_t^\top), \quad \mathbb{E}[G_t \mid \mathcal{H}_{t-1}] = \widetilde{M}_t + \widetilde{B}_t, \quad (3)$$

with bias  $\widetilde{B}_t = -(\delta_\sigma/d) I_d$  exactly on scaled-sphere probes, hence  $\|\widetilde{B}_t\|_{\text{op}} = |\delta_\sigma|/d$  (Lemma B.6). Averaging lifted samples across the probe rounds of segment  $k$  yields  $\widetilde{M}_k := m_k^{-1} \sum_{t \in \mathcal{T}_k} G_t$  (with  $\mathcal{T}_k := \mathcal{T}_{\text{probe}} \cap \mathcal{I}_k$  the probe rounds in segment  $k$ ,  $m_k := |\mathcal{T}_k|$ ), whose top- $r$  eigenvectors form  $\widehat{U}_k \in \mathbb{R}^{d \times r}$  with  $\|\widehat{P}_k - P_k^*\|_{\text{op}} \leq C_{\text{sub}} \sqrt{\log(d/\delta)/m_k}$  (Corollary B.10 below). The variance misspecification bias  $\widetilde{B}$  is a scaled identity (population level), shifting all eigenvalues uniformly and *not rotating eigenvectors*: subspace recovery is robust to the plug-in estimate of  $\sigma_\varepsilon^2$  at the population level. Finite-sample projector concentration

134 still has a floor  $\Delta_\sigma \propto |\hat{\sigma}^2 - \sigma_\varepsilon^2|/(d\lambda_{\min})$  due to the random fluctuation around the biased  
 135 mean (Cor. B.10).

136 Positive identification through the lifted moment  $G_t$  above and the necessity result that  
 137 follows together yield the following theorem, the foundational result of the paper.

138 **Theorem 2.2** (Identification under priced probing). *Consider the model of §2 with scaled-*  
 139 *sphere probes  $u_t = \sqrt{d}v_t$ ,  $v_t \sim \text{Unif}(\mathbb{S}^{d-1})$ , and the lifted probe sample  $G_t = \mathcal{K}^{-1}(s_t u_t u_t^\top)$*   
 140 *from (3).*

141 1. **Identifiability.** *Under (i) known noise variance  $\sigma_\varepsilon^2$  ( $\delta_\sigma = 0$ ), (ii) bounded state-noise*  
 142 *coupling  $\|\mathbb{E}[\varepsilon_t \theta_t]\|_2 \leq \epsilon_\times$ , and (iii) full-dimensional probe support, the lifted sample*  
 143 *satisfies  $\mathbb{E}[G_t | \mathcal{H}_{t-1}] = \widetilde{M}_t$  exactly (the bias  $\widetilde{B}_t = -(\delta_\sigma/d)I_d$  from Lemma B.6 vanishes*  
 144 *under (i); the state-noise coupling term vanishes by sphere symmetry, regardless of  $\epsilon_\times$ ),*  
 145 *and the segment-averaged estimator  $\widehat{M}_k = m_k^{-1} \sum_{t \in \mathcal{T}_k} G_t$  satisfies, with probability at*  
 146 *least  $1 - \delta$ , the projector concentration  $\|\widehat{P}_k - P_k^\star\|_{\text{op}} \leq C_{\text{sub}} \sqrt{\log(d/\delta)/m_k}$ ; in particular*  
 147  *$\widehat{U}_k = \text{TopEig}_r(\widehat{M}_k)$  recovers  $\text{range}(B_k^\star)$  at rate  $1/\sqrt{m_k}$  (Proof in Appendix B.2).*

148 2. **Necessity (unrestricted-second-moment scope).** *Each of (i), (ii), (iii) is individu-*  
 149 *ally necessary in the unrestricted second-moment problem: removing any one yields two*  
 150 *distinct conditional second moments  $\widetilde{M}_t \neq \widetilde{M}_t'$  that generate identical observation laws*  
 151  *$\mathbb{E}[s_t | \mathcal{H}_{t-1}, u]$ , so  $\widetilde{M}_t$  is not identifiable from quadratic measurements. The three con-*  
 152 *structions are independent (each counterexample satisfies the remaining two conditions),*  
 153 *so the three form a minimal identifiability set in this class. Within the rank- $r$  LDS class*  
 154 *of §2 ( $r < d$ ), rank deficiency partially relaxes (i) since  $\sigma_\varepsilon^2$  is identifiable from the  $d - r$*   
 155 *smallest eigenvalues; (ii) and (iii) remain needed (Proof in Appendix B.3).*

### 156 3 Algorithm

157 SPSC (Alg. 1) alternates two phases:

- 158 • **Probe rounds** ( $t \in \mathcal{T}_{\text{probe}}$ ): draw  $u_t \sim Q$  (the probe distribution from §2; e.g. the  
 159 scaled-sphere distribution of Remark 2.1), observe  $y_t$ , accumulate  $G_t$  in the current-  
 160 segment estimator  $\widehat{M}_t$ , and refresh  $\widehat{U}_t = \text{TopEig}_r(\widehat{M}_t)$ .
- 161 • **Exploitation rounds** ( $t \notin \mathcal{T}_{\text{probe}}$ ): project each action  $x \in \mathcal{A}_t$  to  $z_t(x) = \widehat{U}_{t-1}^\top x \in \mathbb{R}^r$ ,  
 162 maintain a *windowed* ridge-UCB estimate  $\widehat{a}_t = \widetilde{V}_t^{-1} \widetilde{b}_t$  over a sliding window  $\mathcal{W}_t$  of the  
 163 last  $W$  exploitation rounds, and play  $x_t = \arg \max_{x \in \mathcal{A}_t} \{z_t(x)^\top \widehat{a}_t + \beta_t^{(r,W)} \|z_t(x)\|_{\widetilde{V}_t^{-1}} +$   
 164  $\gamma_t \|x\|_2\}$ .

165 The confidence radius  $\beta_t^{(r,W)} = \sigma_\varepsilon \sqrt{r \log(1 + WR_A^2/(\lambda r)) + 2 \log(2K/\delta)} + \sqrt{\lambda} S_w$  scales with  
 166  $\sqrt{r}$ , not  $\sqrt{d}$ : this is where the intrinsic-rank improvement enters. The correction term  
 167  $\gamma_t = \Gamma_k + V_{k,t}(W)$  absorbs subspace-mismatch error  $\Gamma_k \propto \varepsilon_k := \|\widehat{P}_k - P_k^\star\|_{\text{op}}$  and local within-  
 168 window drift  $V_{k,t}(W) := \sum_{s, s+1 \in \mathcal{I}_k, t-W \leq s < t} \|\theta_{s+1} - \theta_s\|_2$  (formal definitions in App. B.4).  
 169 Per-round cost is  $O(dr|\mathcal{A}_t| + r^2)$  vs.  $O(d^2|\mathcal{A}_t|)$  for ambient LinUCB, an  $O(d/r)$  speed-up in  
 170 addition to the statistical gain.

171 Full pseudocode for Alg. 1 (oracle boundaries) is in Appendix D, alongside the adaptive  
 172 variant described next.

173 **Adaptive variant (unknown boundaries).** SPSC-Adaptive removes the oracle-  
 174 boundary assumption by detecting segment changes online: it maintains two non-  
 175 overlapping rolling-window estimators  $\widehat{M}_t^{\text{recent}}, \widehat{M}_t^{\text{past}}$  of the probe-time second moment and  
 176 triggers a subspace reset whenever  $S_t := \|\widehat{M}_t^{\text{recent}} - \widehat{M}_t^{\text{past}}\|_{\text{op}}$  exceeds a calibrated threshold  
 177  $b$ . The threshold is chosen so that, with probability  $1 - \delta_{\text{FA}}$ , no false alarm occurs across the  
 178 horizon, while every true change of size  $\Delta_k \geq 2b$  is detected within a bounded delay  $D_{\text{max}}$   
 179 (specified in Appendix D together with the full pseudocode, tuning, and CUSUM analysis).

## 4 Regret guarantees

**Balanced probe schedule.** The probe schedule is *balanced*: there is an absolute constant  $a_0$  such that, for every segment  $k \in \{1, \dots, K\}$  and every prefix-count  $j \in \{0, 1, \dots, m_k\}$ , the number of exploit rounds in  $E_k$  with exactly  $j$  probes collected before them is at most  $a_0 \lceil \ell_k / m_k \rceil$ . (Uniformly spaced probes satisfy this.)

**Theorem 4.1** (Costed dynamic regret of Alg. 1). *Run Alg. 1 with oracle segment boundaries  $\{\mathcal{I}_k\}_{k=1}^K$ , scaled-sphere probes, and a balanced probe schedule. Under the model of §2, Assumption B.2, and exact centering  $\hat{\sigma}^2 = \sigma_\varepsilon^2$ , set the per-segment probe budget to  $m_k = \min\{\ell_k, \lceil c_0 \ell_k^{2/3} \rceil\}$  for an absolute constant  $c_0 > 0$  (chosen as the closed-form optimum (13); segments shorter than the burn-in threshold  $q^* = O(\log(KdT/\delta)/\lambda_{\min}^2)$  are charged trivially), where  $\lambda_{\min} > 0$  is the predictable probe-time  $r$ th-eigenvalue lower bound (Lem. B.1). Then with probability at least  $1 - \delta$  and for fixed  $K$ ,*

$$\text{DynReg}_T^{(c)} \leq \tilde{O}(r\sqrt{T}) + \tilde{O}(T^{2/3}) + O(WV_{\text{in}}). \quad (4)$$

*Including all constants:  $\text{DynReg}_T^{(c)} \leq \tilde{O}(r\sqrt{KT}) + \tilde{O}(A^{1/3}B^{2/3}K^{1/3}T^{2/3}) + O(R_A W V_{\text{in}}) + \tilde{O}(K^{2/3}T^{1/3}/\lambda_{\min}^2)$ , with  $A := (R_A + R_Q)S_w + c$ ,  $B := CR_A S_w(1 + R_A \sqrt{W/\lambda})\sqrt{\log(2KdT/\delta)}$ ,  $R_Q := \sup_{u \in \text{supp}(Q)} \|u\|_2$  ( $R_Q = \sqrt{d}$  for scaled-sphere;  $R_Q \leq R_A$  if probes lie inside the action set); the  $\tilde{O}(K^{2/3}T^{1/3}/\lambda_{\min}^2)$  summand is the burn-in cost (lower order than  $\tilde{O}(T^{2/3})$  for fixed  $K$ ). Here  $V_{\text{in}} := \sum_k \sum_{s,s+1 \in \mathcal{I}_k} \|\theta_{s+1} - \theta_s\|_2$  is the within-segment path variation and  $W$  is the exploitation window (cumulative on  $E_k$  when  $W \geq \ell_k$ ); the bound is most informative when  $V_{\text{in}}$  is small. Proof in §B.5.*

**Corollary 4.2** (Plug-in variance). *Estimate  $\hat{\sigma}^2$  from  $N$  probe-round residuals (Lem. B.11) and write  $\delta_\sigma := |\hat{\sigma}^2 - \sigma_\varepsilon^2|$ . For scaled-sphere probes, the bias  $\tilde{B} = -(\delta_\sigma/d)I_d$  from Lem. B.6 is a scaled identity, hence preserves the eigenvectors of  $\bar{M}_k^{\text{probe}}$ . Davis–Kahan applied with the biased mean as reference removes the  $\Delta_\sigma$  floor at population level, so Theorem 4.1 holds with the displayed bound,  $\lambda_{\min}$  replaced by  $\lambda_{\min} - \delta_\sigma/d$  in the constants. Choosing  $N \gtrsim T^{2/3}/d^2$  yields  $\delta_\sigma/d \ll \lambda_{\min}$  with high probability, so the leading rate is preserved.*

**Corollary 4.3** (Within-segment-stationary special case). *Suppose  $V_{\text{in}} = 0$ , i.e. within each segment  $\mathcal{I}_k$  the parameter is constant ( $\theta_t \equiv \theta^{(k)}$  for  $t \in \mathcal{I}_k$ , with  $\theta^{(k)} \in \text{range}(B_k^*)$ ). The non-degenerate innovation assumption ( $\Sigma_{\eta,k} \succ 0$ ) still holds across the segment family, so the rank- $r$  subspace  $\text{range}(B_k^*)$  is well-defined and identifiable through SPSC’s probe channel. The bound of Theorem 4.1 reduces to*

$$\text{DynReg}_T^{(c)} \leq \tilde{O}(r\sqrt{T}) + \tilde{O}(T^{2/3})$$

*for fixed  $K$  (suppressing polylog factors). The leading  $\tilde{O}(r\sqrt{T})$  term comes from LinUCB on the  $r$ -dimensional projected target  $z_t^\top a^{(k)}$  (this does not require the within-segment target to itself span an  $r$ -dimensional set; the projector is what collapses the geometry to dimension  $r$ , not the trajectory of  $\theta_t$ ). This is the regime that class- or cohort-induced segment shifts approximate empirically.*

**Scope of the bound.** Theorem 4.1 analyzes exactly the practical Alg. 1: probes are interleaved according to a balanced schedule,  $\hat{U}_t$  is refreshed online, and exploitation rounds re-project the windowed history through the current basis. The time-varying-basis subspace error  $\varepsilon_{k,t} = \|\hat{P}_t - P_k^*\|_{\text{op}}$  is controlled uniformly in  $t$  via a prefix concentration argument (Lemma B.17).

The leading  $\tilde{O}(r\sqrt{T})$  term beats the ambient baseline  $\tilde{O}(d\sqrt{T})$  whenever  $r \ll d$ ; the  $\tilde{O}(T^{2/3})$  term is the price of acquiring the subspace information online. The constant  $B$  depends on the exploitation window  $W$  as  $\sqrt{W}$ ; for  $W$  treated as a problem parameter (the setting throughout this paper, consistent with  $W \leq \min_k \ell_k$  on  $\ell_k$ -bounded segments), this is absorbed into the constants and the  $\tilde{O}(T^{2/3})$  rate is preserved. If  $W$  is allowed to scale with  $T$ , the second term becomes  $\tilde{O}(W^{1/3}T^{2/3})$ . Solving  $r\sqrt{T} \lesssim T^{2/3}$  gives a simple rule of thumb: SPSC tends to outperform ambient LinUCB whenever  $d - r \gtrsim T^{1/6}$ . The experiments in §5 show a crossover qualitatively consistent with this rule of thumb.

228 **Corollary 4.4** (Rank-adaptive guarantee). *If the true rank  $r$  is unknown, thresholding  $\widehat{M}_k$*   
 229 *at  $\tau_k^{\text{rank}} := 2R_X \sqrt{\log(2d/\delta)/m_k}$  ( $R_X$  the lifted-sample envelope from App. B.2) recovers the*  
 230 *true rank with probability  $\geq 1 - \delta$  whenever the population eigengap exceeds  $4\tau_k^{\text{rank}}$ , and the*  
 231 *bound (4) applies unchanged. Proof in §B.6.*

232 **Proposition 4.5** (Adaptive SPSC). *Suppose every true change satisfies  $\Delta_k \geq 2b$  for the*  
 233 *threshold  $b$  set in Appendix B.7. Then with probability at least  $1 - \delta_{\text{FA}}$ , SPSC-Adaptive*  
 234 *(Alg. 2) attains the same leading rate as Theorem 4.1 with an additive  $O(KW_{\text{det}})$  delay*  
 235 *overhead. Proof in Appendix B.7.*

## 236 5 Experiments

237 We evaluate SPSC and SPSC-Adaptive on eleven benchmarks: a synthetic phase-transition  
 238 grid; UCI/MovieLens (Covertime, Pendigits, Satimage, MNIST, Fashion-MNIST, Movie-  
 239 Lens); clinical (Warfarin, Vancomycin); the Russac small- $d$  stress test (§5.4); and Open  
 240 Bandit production logs (§5.5). Baselines: LinUCB [Abbasi-Yadkori et al., 2011], D-  
 241 LinUCB [Russac et al., 2019], SW-LinUCB [Cheung et al., 2019], Restart-LinUCB,  
 242 LowRank-Reward, LowOFUL [Jun et al., 2019], VOFUL [Kim et al., 2022], LinTS, SW-  
 243 LinTS, and adapted BOSS [Duong et al., 2024]/Jedra [Jedra et al., 2024]; Oracle-LinUCB  
 244 (true subspace) is an optimistic reference. Curves are means over  $\geq 10$  seeds with  $\pm 1$  SE  
 245 bands.

### 246 5.1 Phase-transition boundary

247 Theory predicts a crossover at  $d - r \asymp T^{1/6}$ : SPSC wins above it, ambient Lin-  
 248 UCB below. We verify on a piecewise low-rank LDS at  $T=5,000$ , sweeping  $(d, r) \in$   
 249  $\{5, 10, 20, 30, 45, 60, 80, 100\} \times \{1, 3, 5, 10, 15, 20\}$  with  $r < d$  (40 cells),  $K=10$ ,  $\sigma_\varepsilon=0.3$ , spectral  
 250 radius 0.99, 40 actions, probe cost  $c=0.1$ .

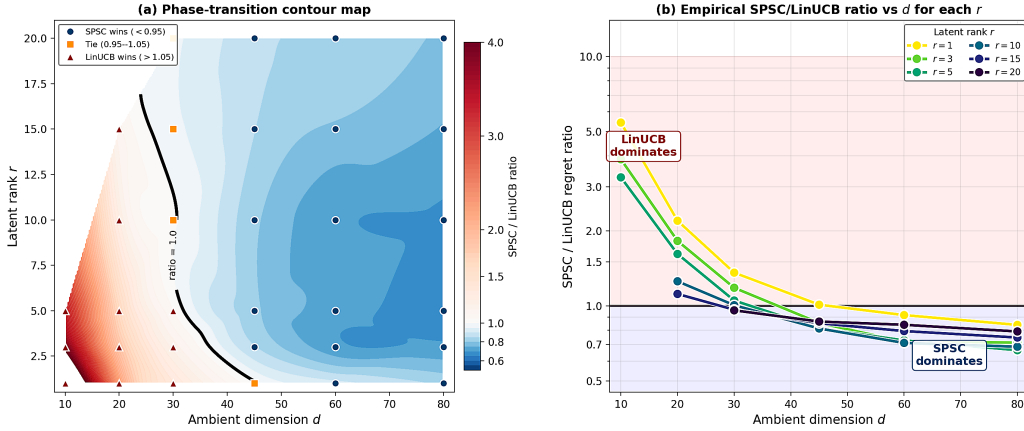


Figure 1: **Phase-transition boundary.** SPSC/LinUCB costed-regret ratio across the  $(d, r)$  grid. The crossover  $d - r = T^{1/6}$  (dashed) separates SPSC-wins (ratio  $< 1$ ) from ambient-wins; per-rank curves track  $\sqrt{r/d}$  (dotted), matching Theorem 4.1.

251 SPSC achieves 13–34% reductions whenever  $d \geq 45$  with every  $r$ -curve tracking  $\sqrt{r/d}$ : use  
 252 SPSC when  $d - r \gtrsim T^{1/6}$ , and ambient LinUCB otherwise.

### 253 5.2 Real-data multi-baseline comparison

254 Six real-data benchmarks recast as piecewise low-rank bandits on a  $(d, r)$ -grid: UCI **Cover-**  
 255 **type** ( $K=4$ ,  $T=10,000$ ); **Pendigits**, **Satimage**, pixelized **MNIST** and **Fashion-MNIST**  
 256 (random-projection features), and **MovieLens-100K** with genuine user ratings (all  $K=10$ ,  
 257  $T=5,000$ , 10 seeds). Segments are class or user-subset shifts. The first five have cluster-  
 258 induced low-rank structure; MovieLens is only approximately low-rank (rank- $d$  SVD fea-

tures, §E) and serves as a robustness stress test. Table 1 reports SPSC vs. LinUCB and vs. best non-oracle competitor on representative cells; per-cell tables in Appendix F.

Table 1: **SPSC/LinUCB** and **SPSC-Adaptive/Best Competitor** ratios across six real-data benchmarks (10 seeds per cell). Values  $<1$  indicate SPSC wins; **bold** marks cells where an SPSC variant beats every non-oracle baseline. SPSC trails LinUCB at  $d \approx r$  (top of each block) and gains as  $d$  grows relative to  $r$ . Restart-LinUCB, the closest competitor, is subspace-oblivious, so its regret is constant along each row.

| Dataset / $r$         | $d=55$      |             | $d=105$     |             | $d=155/200$      |             | best      |             |
|-----------------------|-------------|-------------|-------------|-------------|------------------|-------------|-----------|-------------|
|                       | S/Lin       | S/Best      | S/Lin       | S/Best      | S/Lin            | S/Best      | cell      | ratio       |
| Covertypes, $r=10$    | 0.96        | 0.99        | <b>0.88</b> | <b>0.92</b> | —                | —           | (105, 20) | <b>0.86</b> |
| Pendigits, $r=10$     | <b>0.76</b> | <b>0.77</b> | <b>0.65</b> | <b>0.73</b> | —                | —           | (105, 10) | <b>0.65</b> |
| Satimage, $r=10$      | <b>0.47</b> | <b>0.59</b> | <b>0.40</b> | <b>0.58</b> | —                | —           | (105, 10) | <b>0.40</b> |
| MNIST, $r=10$         | <b>0.84</b> | <b>0.86</b> | <b>0.74</b> | <b>0.88</b> | <b>0.73/0.98</b> | <b>0.98</b> | (105, 20) | <b>0.74</b> |
| Fashion-MNIST, $r=10$ | <b>0.88</b> | <b>0.90</b> | <b>0.73</b> | <b>0.88</b> | <b>0.67/0.99</b> | 0.99        | (105, 20) | <b>0.71</b> |
| MovieLens, $r=10$     | 1.00        | 1.00        | <b>0.97</b> | <b>0.97</b> | <b>0.92/0.93</b> | <b>0.93</b> | (200, 5)  | <b>0.92</b> |

At ( $d=105, r=10$ ), SPSC-Adaptive beats LinUCB by  $-47\%$  on Pendigits (operating-regime grid in Fig. 2, App. F) and  $-68\%$  on Satimage (closest competitor VOFUL is  $+112\%$  above SPSC-Adaptive; per-cell results in Table 10, App. F); at ( $d=105, r=20$ ) on Covertypes it beats Restart-LinUCB by  $-15\%$ . Non-low-rank baselines are by construction  $r$ -insensitive.

### 5.3 Clinical benchmarks: Warfarin and Vancomycin dosing

**Warfarin.** Warfarin dosing depends on demographics, genotype (VKORC1, CYP2C9), and clinical indicators [Consortium, 2009]. We calibrate to the IWPC cohort ( $\sim 5,700$  patients,  $d=93$ ,  $K=8$  segments,  $T=5,000$ , 10 seeds; reward = negative deviation from therapeutic dose). Table 2 ( $r=3$ ): SPSC-Adaptive cuts regret by 67.3% vs. LinUCB; full per-rank tables in Table 14 (App. F); random-subspace ablation in App. I.

**Vancomycin.** With the AUC-targeted Rybak et al. [2020] protocol (Cockcroft–Gault  $\text{CrCl} \rightarrow \text{CL}_{\text{vanco}} = 0.04 \text{CrCl}$ , daily dose at  $\text{AUC}_{24}=500$ ;  $d=93$ ,  $K=8$ ,  $T=5,000$ , 10 seeds), Table 3 sweeps  $r \in \{1, 2, 3, 5, 10\}$ : SPSC-Adaptive cuts regret by 36.4% at  $r=1$  and 14.5–25.7% for  $r \in \{2, 3, 5, 10\}$  vs. LinUCB, beating every non-oracle baseline at every rank. Per-rank tables in Appendix K.

### 5.4 Standard nonstationary benchmark

On the small- $d$  stress test of Russac et al. [2019] ( $d=2$ ,  $r=1$ ,  $K=4$ ,  $T=6,000$ , 50 arms, 30 seeds), SPSC reduces regret by  $-46.5\%$  vs. D-LinUCB,  $-34.8\%$  vs. SW-LinUCB, and  $-52.0\%$  vs. stationary OFUL (itself  $+11.5\%$  above D-LinUCB here). Full table in Appendix H.

### 5.5 Real exploration logs: Open Bandit Pipeline (ZOZO)

The Open Bandit Dataset [Saito et al., 2021] ships production exploration logs (random-policy ZOZOTOWN slice,  $T=5,000$ ,  $K=10$ , 10 seeds); only approximately low-rank, so this is a *robustness* stress test rather than a setting where the rank- $r$  assumption holds exactly. At  $d=55$ ,  $r=5$ , SPSC reaches  $83 \pm 2$  vs. LinUCB’s  $102 \pm 1$  ( $-18.8\%$ ); SPSC-Adaptive  $-10.9\%$ . SPSC neither breaks nor over-claims on weakly-low-rank production data. Per-cell results in Table 15 (App. F).

### 5.6 Comparison with stationary low-rank methods (oracle restarts)

For a best-effort comparison with stationary low-rank methods, we adapt BOSS [Duong et al., 2024] and Jedra [Jedra et al., 2024] by restarting at every true segment boundary, an oracle advantage we do not give SPSC (setup in Appendix J). Table 4: SPSC wins all eight cells by 5.8–19.9%, with the lead largest at small  $d$  ( $-16.3$  to  $-19.9\%$  at  $d=55$ ) and narrowing to 5.8–8.0% at  $d=200$ . SPSC-Adaptive (no oracle) also beats both on every cell.

Table 2: **Warfarin** (representative cell  $r=3$ , mean $\pm$ SE, 10 seeds). SPSC-Adaptive cuts regret by 67.3% vs. LinUCB. Full per-rank tables in Appendix F.

| Method                      | Costed regret                  | vs. LinUCB    |
|-----------------------------|--------------------------------|---------------|
| Oracle LinUCB               | 177 $\pm$ 4                    | −91.2%        |
| <b>SPSC Alg. 1 (ours)</b>   | 1372 $\pm$ 33                  | −31.6%        |
| <b>SPSC-Adaptive (ours)</b> | <b>657 <math>\pm</math> 39</b> | <b>−67.3%</b> |
| LowOFUL                     | 683 $\pm$ 97                   | −66.0%        |
| VOFUL                       | 779 $\pm$ 93                   | −61.2%        |
| LowRank-Reward              | 1056 $\pm$ 40                  | −47.4%        |
| SW-LinUCB                   | 2096 $\pm$ 18                  | +4.5%         |
| LinUCB                      | 2006 $\pm$ 17                  | —             |

Table 3: **Vancomycin** (mean $\pm$ SE, 10 seeds): SPSC-Adaptive control regret across range of  $r$  vs. LinUCB and the best non-oracle baseline. Full tables in Appendix K.

| $r$ | SPSC-Adaptive                      | vs. LinUCB | vs. best comp. |
|-----|------------------------------------|------------|----------------|
| 1   | <b>525.3 <math>\pm</math> 45.8</b> | −36.4%     | −35.1%         |
| 2   | <b>613.6 <math>\pm</math> 33.1</b> | −25.7%     | −17.1%         |
| 3   | <b>640.3 <math>\pm</math> 33.0</b> | −22.4%     | −10.1%         |
| 5   | <b>665.9 <math>\pm</math> 27.2</b> | −19.3%     | −12.7%         |
| 10  | <b>705.6 <math>\pm</math> 19.5</b> | −14.5%     | −7.0%          |

Table 4: **Adapted stationary low-rank methods on a  $(d, r)$ -grid.** Costed regret (mean $\pm$ SE, 10 seeds,  $T=5,000$ ,  $K=10$ ). BOSS/Jedra restart at *oracle* segment boundaries; SPSC-Adaptive detects them online. SPSC variants win every cell.

| $d$ | $r$ | SPSC Alg. 1                    | SPSC-Adaptive | BOSS (oracle) | Jedra (oracle) | SPSC vs. best comp. |
|-----|-----|--------------------------------|---------------|---------------|----------------|---------------------|
| 55  | 5   | <b>323 <math>\pm</math> 10</b> | 342 $\pm$ 15  | 399 $\pm$ 17  | 386 $\pm$ 18   | −16.3%              |
| 55  | 10  | <b>471 <math>\pm</math> 10</b> | 507 $\pm$ 13  | 588 $\pm$ 27  | 602 $\pm$ 24   | −19.9%              |
| 105 | 5   | <b>264 <math>\pm</math> 9</b>  | 274 $\pm$ 10  | 300 $\pm$ 14  | 299 $\pm$ 13   | −11.7%              |
| 105 | 10  | <b>377 <math>\pm</math> 15</b> | 401 $\pm$ 14  | 423 $\pm$ 18  | 429 $\pm$ 14   | −10.9%              |
| 105 | 20  | <b>578 <math>\pm</math> 12</b> | 608 $\pm$ 14  | 654 $\pm$ 17  | 654 $\pm$ 14   | −11.6%              |
| 200 | 5   | <b>208 <math>\pm</math> 5</b>  | 218 $\pm$ 6   | 236 $\pm$ 9   | 226 $\pm$ 9    | −8.0%               |
| 200 | 10  | <b>277 <math>\pm</math> 10</b> | 292 $\pm$ 12  | 294 $\pm$ 11  | 305 $\pm$ 15   | −5.8%               |
| 200 | 20  | <b>442 <math>\pm</math> 13</b> | 452 $\pm$ 11  | 473 $\pm$ 16  | 489 $\pm$ 18   | −6.6%               |

294 **Robustness of SPSC-Adaptive.** Detailed comparison under correct vs. misspecified  
 295 oracle boundaries is in Appendix L: SPSC-Adaptive is robust to false-alarm boundaries,  
 296 while Alg. 1 degrades monotonically as the oracle is corrupted.

## 297 5.7 Rank misspecification, probe-rate sensitivity, and take-away

298 **Rank misspecification (Covertime,  $d=155$ ,  $r^*=10$ ; App. G.2):** sweeping over nine  
 299 grid points  $r \in \{1, 3, 5, 10, 15, 20, 30, 50, 80\}$  gives a U-shaped curve (under: +33% at  $r=1$ ;  
 300 sweet spot  $r=30$ , −10%); SPSC beats LinUCB across  $r \in [15, 50]$ , so mild overestimation  
 301 is safe. **Probe rate** (App. G.1): on the synthetic reference setting ( $d=4$ ,  $r=1$ ), the probe-  
 302 period sweep  $\{5, 10, 20, 30, 50, 100, 300\}$  has its optimum near 20–30, matching  $m_k^* \propto \ell_k^{2/3}$ .  
 303 **Additional datasets** (see Appendix F): SPSC wins every cell at  $d \geq 105$ , peaking at  
 304  $(d, r)=(105, 20)$ . **Sensitivity & dynamics.** Large-scale assumption sweeps (App. G.4)  
 305 confirm SPSC tracks Oracle within a constant factor through variance misspecification,  
 306 approximate-rank perturbation ( $\epsilon_k^\perp$  up to 0.4), and spectral radius up to  $\rho=0.95$ ; SPSC  
 307 remains the best non-Oracle method at  $\rho=0.99$ . Subspace error tracks the predicted  $1/\sqrt{m}$   
 308 rate (App. G.5); noise/change-point-frequency/drift-speed sweeps (App. G.6) confirm the  
 309 same trends. **Take-away.** Across the eleven benchmarks, SPSC delivers  $\sqrt{r/d}$  intrinsic-  
 310 rank savings whenever  $d - r \gtrsim T^{1/6}$  and does not catastrophically fail outside that regime.

## 6 Conclusion

Sequential decision problems in recommendation, clinical dosing, and adaptive experimentation share two features: rewards live on a low-rank subspace of rank  $r \ll d$ , and that subspace shifts at unknown change points. Stationary low-rank, nonstationary, and cost-aware bandits each cover one piece; none recover a changing subspace under priced single-play scalar feedback. We close this gap: three probe-side conditions, known noise variance, bounded state-noise coupling, and full-dimensional probe support, are each individually necessary (in the unrestricted-second-moment scope) and jointly sufficient for subspace recovery from quadratic functionals of scalar rewards (Theorem 2.2). Based on this, we propose SPSC, which interleaves probing with windowed projected ridge-UCB in the learned  $r$ -dimensional subspace and achieves costed dynamic regret  $\tilde{O}(r\sqrt{T}) + \tilde{O}(T^{2/3}) + O(WV_{\text{in}})$ , replacing the ambient  $d\sqrt{T}$  rate with the intrinsic  $r$  (Theorem 4.1); a CUSUM adaptive variant attains the same rate without oracle change points (Proposition 4.5). On eleven benchmarks across synthetic, UCI, clinical, and production data, SPSC reduces regret whenever  $d - r \gtrsim T^{1/6}$ , a crossover consistent with theory.

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## 421 A Comparison with prior work

422 This appendix positions SPSC against published work in low-rank bandits, nonstationary  
423 bandits, and active-query bandits via three tables: Table 5 (head-to-head with seven recent  
424 works on regret form, change-point adaptation, and probing budget), Table 6 (compact po-  
425 sitioning of SPSC’s combined setting against the three closest *problem classes*), and Table 7  
426 (regret-rate comparison with five published baselines that appear in our experiments). The  
427 take-away is that the existing literature splits along three axes (low-rank, nonstationary,  
428 costly probing) and no prior method addresses all three; SPSC is the first to do so with  
429 a self-contained identification chain that turns rank-one scalar rewards into matrix-valued  
430 evidence.

## 431 B Proofs

432 This section proves the statements of §4 in dependency order:

- 433 1. §B.1 fixes notation and establishes the probe-time excitation lemma that all proofs de-  
434 pend on.
- 435 2. §B.2 derives the load-bearing *identification chain*: closed-form inverse of the probe oper-  
436 ator  $\mathcal{K}$ , quadratic-measurement identity, lifted probe sample’s near-unbiasedness, matrix  
437 Bernstein/Freedman, and the projector error via Davis–Kahan.
- 438 3. §B.4 states and proves the interleaved-analysis lemmas: projected nonstationary UCB  
439 with time-varying basis (Lem. B.16), uniform prefix subspace concentration (Lem. B.17),  
440 and cumulative interleaved subspace error (Lem. B.18).

Table 5: Comparison with related literature. “Prob.” indicates a probabilistic regret bound; “CP?” indicates change-point adaptation.

| Reference                    | Venue/<br>Year | Regret / Theory                                    | Prob. | CP? | Difference<br>vs. ours               |
|------------------------------|----------------|----------------------------------------------------|-------|-----|--------------------------------------|
| Kang et al. [2022]           | NeurIPS’22     | $\tilde{O}\left(\sqrt{(d_1 + d_2)MrT}\right)$      | ✓     | ✗   | Stationary; probing not budgeted.    |
| Stojanovic et al. [2023]     | NeurIPS’23     | Entry-wise / subspace recovery                     | ✗/✓   | ✗   | Estimation focus; no probe budget.   |
| Jedra et al. [2024]          | ICML’24        | $\tilde{O}\left(r^{5/4}(m+n)^{3/4}\sqrt{T}\right)$ | ✓     | ✗   | Stationary; no re-learning.          |
| Jang et al. [2024]           | ICML’24        | Arm-set-dependent regret                           | ✓     | ✗   | Stationary; no CP detection.         |
| Abbasi-Yadkori et al. [2023] | JMLR’23        | $\tilde{O}(\sqrt{KN(S+1)})$                        | ✗     | ✓   | Unstructured $K$ -arm; no low-rank.  |
| Komiyama et al. [2024]       | JMLR’24        | ADR-bandit finite-time                             | ✗     | ✓   | Unstructured; no low-rank + probing. |
| Hou et al. [2024]            | NeurIPS’24     | Fixed-confidence sample complexity                 | ✓     | ✓   | BAI objective; no low-rank.          |

Table 6: Compact positioning of SPSC’s combined setting.

| Work class                                | Low-rank | Nonstationary | Costly probing | Main gap vs. ours                                                                                                                                      |
|-------------------------------------------|----------|---------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low-rank bandits                          | ✓        | ✗             | ✗              | Fixed subspace; no piecewise changes; no sensing-cost budget.                                                                                          |
| Nonstationary linear/contextual bandits   | ✗        | ✓             | ✗              | Adapt in ambient parameter space; ignore latent low-rank structure.                                                                                    |
| Costly-information / active-query bandits | ✗        | ✗             | ✓              | Price information acquisition; do not address subspace recovery under changing representations.                                                        |
| <b>This paper</b>                         | ✓        | ✓             | ✓              | <b>Combines low-rank structure, piecewise-stationary adaptation, and explicit costly probing in a single identification-to-exploitation framework.</b> |

Table 7: Regret-scaling comparison with the published baselines that appear in our experiments.

| Algorithm          | Reference                    | Regret scaling                          | Key difference vs. SPSC                         |
|--------------------|------------------------------|-----------------------------------------|-------------------------------------------------|
| LinUCB             | Abbasi-Yadkori et al. [2011] | $O(d\sqrt{T})$                          | Ambient; no subspace exploitation.              |
| LowOFUL            | Jun et al. [2019]            | $\tilde{O}(r\sqrt{T})$                  | Needs oracle subspace $B$ .                     |
| LowRank-Reward     | Lu et al. [2021]             | $\tilde{O}(r\sqrt{T})$                  | Stationary; learns $B$ from rewards.            |
| BOSS               | Duong et al. [2024]          | $\tilde{O}(r\sqrt{dT})$                 | Stationary; no online change adaptation.        |
| Jedra              | Jedra et al. [2024]          | $\tilde{O}(r^{5/4}(m+n)^{3/4}\sqrt{T})$ | Stationary; no probe cost.                      |
| <b>SPSC (ours)</b> | This work                    | $\tilde{O}(r\sqrt{T} + T^{2/3})$        | Learns a <i>changing</i> $B$ via priced probes. |

441 4. §B.5 proves the main costed regret bound (Thm. 4.1) via a five-step decomposition: probe  
 442 regret, exploitation regret via Lem. B.16, per-segment bound, probe-estimation tradeoff  
 443 with burn-in, and sum over segments.  
 444 5. §B.6 derives the rank-adaptive corollary (Cor. 4.4) via Weyl’s inequality on the matrix-  
 445 Bernstein bound.  
 446 6. §B.7 lifts the guarantee to SPSC-Adaptive (Prop. 4.5) by controlling the CUSUM detec-  
 447 tor.  
 448 The windowed projected self-normalized bound (Lem. B.3), the windowed projected poten-  
 449 tial condition (Assumption B.2), and the projected nonstationary UCB lemma (Lem. B.16)  
 450 are the linear-bandit ingredients; constants that do not affect leading order are absorbed  
 451 into  $\tilde{O}(\cdot)$ .

## 452 B.1 Preliminaries and notation

453 Throughout,  $\mathcal{I}_k = [\tau_{k-1}, \tau_k)$  denotes segment  $k$  of length  $\ell_k := |\mathcal{I}_k|$ , so  $\sum_k \ell_k = T$ . Inside  
 454  $\mathcal{I}_k$  the learner performs  $m_k = |\mathcal{T}_k|$  probe rounds ( $\mathcal{T}_k := \mathcal{T}_{\text{probe}} \cap \mathcal{I}_k$ ) and  $n_k = \ell_k - m_k$   
 455 exploitation rounds ( $E_k := \mathcal{I}_k \setminus \mathcal{T}_k$ ). We write  $\hat{P}_t = \hat{U}_t \hat{U}_t^\top$  and  $P_k^* = B_k^* B_k^{*\top}$  for the estimated  
 456 and true rank- $r$  projectors, and  $\varepsilon_k := \|\hat{P}_k - P_k^*\|_{\text{op}}$ . We work on the high-probability event  
 457  $\mathcal{G}_w := \{\max_{t \leq T} \|w_t\|_2 \leq S_w\}$ . For sub-Gaussian innovations one may take  $S_w = \tilde{O}(\sqrt{r})$   
 458 via union bound (the failure probability is absorbed into the overall  $\delta$  budget); for bounded  
 459 innovations  $\mathcal{G}_w$  holds deterministically. In what follows we implicitly condition on  $\mathcal{G}_w$ .

460 The *predictable second moment* is  $\tilde{M}_t := \mathbb{E}[\theta_t \theta_t^\top \mid \mathcal{H}_{t-1}] = B_k^* \mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}] (B_k^*)^\top$ , and  
 461 the probe-time average is  $\bar{M}_k^{\text{probe}} := m_k^{-1} \sum_{t \in \mathcal{T}_k} \tilde{M}_t$ . By the non-degenerate-innovation  
 462 assumption  $\Sigma_{\eta,k} \succ 0$  of §2,  $\bar{M}_k^{\text{probe}}$  has rank  $r$  and its  $r$ -th eigenvalue  $\lambda_{\min} > 0$  is uniformly  
 463 bounded below in  $k$  (Lem. B.1 below; stability of  $A_k$  is used only to bound  $\|w_t\| \leq S_w$ , not  
 464 for this lower bound). Constants  $C, C', \dots$  depend only on  $\sigma_\varepsilon, R_{\mathcal{A}}, L, \rho_Q, S_w, \lambda_{\min}$  and may  
 465 change line to line.

466 **Lemma B.1** (Predictable probe-time excitation). *For each segment  $k$ ,*

$$\lambda_r \left( \frac{1}{m_k} \sum_{t \in \mathcal{T}_k} \mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}] \right) \geq \lambda_{\min} > 0 \quad \text{almost surely.}$$

467 *Proof.* For each  $t \in \mathcal{I}_k$ , the conditional second moment satisfies  $\mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}] =$   
 468  $A_k w_{t-1} w_{t-1}^\top A_k^\top + \Sigma_{\eta,k} \succeq \Sigma_{\eta,k}$  (LDS recursion + non-degenerate innovations). Hence  
 469  $\lambda_r(\mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}]) \geq \lambda_r(\Sigma_{\eta,k}) =: \lambda_{\min} > 0$  uniformly across rounds. Averaging over any  
 470 probe subset  $\mathcal{T}_k \subseteq \mathcal{I}_k$  preserves this PSD lower bound:  $\lambda_r(m_k^{-1} \sum_{t \in \mathcal{T}_k} \mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}]) \geq$   
 471  $\lambda_{\min}$  almost surely. Stability of  $A_k$  is therefore not needed for the lemma; the innovation  
 472 covariance alone suffices.  $\square$

473 **Notation for the interleaved analysis.** For each segment  $\mathcal{I}_k = [\tau_{k-1}, \tau_k)$ , recall  $\ell_k =$   
 474  $|\mathcal{I}_k|$ ,  $\mathcal{T}_k = \mathcal{T}_{\text{probe}} \cap \mathcal{I}_k$ ,  $m_k = |\mathcal{T}_k|$ ,  $E_k = \mathcal{I}_k \setminus \mathcal{T}_k$ ,  $n_k = |E_k| = \ell_k - m_k$ . For  $t \in \mathcal{I}_k$ ,  
 475 let  $q_k(t) := |\mathcal{T}_k \cap [\tau_{k-1}, t)|$  be the number of probes collected in segment  $k$  strictly before  
 476 round  $t$ . Denote  $P_k^* = B_k^* B_k^{*\top}$ ,  $\hat{P}_t = \hat{U}_t \hat{U}_t^\top$ , and  $\varepsilon_{k,t} := \|\hat{P}_t - P_k^*\|_{\text{op}}$ . The within-segment  
 477 variation is  $V_k^{\text{in}} := \sum_{s, s+1 \in \mathcal{I}_k} \|\theta_{s+1} - \theta_s\|_2$ ,  $V_{\text{in}} := \sum_k V_k^{\text{in}}$ . Let  $R_Q := \sup_{u \in \text{supp}(Q)} \|u\|_2$   
 478 ( $R_Q = \sqrt{d}$  for scaled-sphere probes;  $R_Q \leq R_{\mathcal{A}}$  if probes are feasible actions covered by the  
 479 action-radius bound).

480 **Balanced interleaved probe schedule.** The probe schedule is balanced: there exists  
 481 an absolute constant  $a_0$  such that, for every segment  $k$  and every  $j \in \{0, 1, \dots, m_k\}$ ,

$$|\{t \in E_k : q_k(t) = j\}| \leq a_0 \lceil \ell_k / m_k \rceil. \quad (\text{B})$$

482 Uniformly spaced probes satisfy this condition.

**Windowed projected potential.** For each exploit round  $t \in E_k$ , Alg. 1 uses the *current* basis  $\widehat{U}_{t-1}$  and re-projects the windowed history through this same basis:  $z_t(x) := \widehat{U}_{t-1}^\top x$ ,  $\widetilde{V}_t := \lambda I_r + \sum_{s \in \mathcal{W}_t} z_t(x_s) z_t(x_s)^\top$ ,  $w_t(x) := \|z_t(x)\|_{\widetilde{V}_t^{-1}}$ .

**Assumption B.2** (Windowed projected potential). *For each segment  $\mathcal{I}_k$ ,*

$$\sum_{t \in E_k} \min\{1, w_t(x_t)^2\} \leq C_{\text{pot}} (n_k/W + 1) r \log\left(1 + \frac{WR_{\mathcal{A}}^2}{\lambda r}\right), \quad (5)$$

*with the convention that the factor  $(n_k/W + 1)$  is interpreted as 1 and  $W$  is replaced by  $n_k$  when  $W \geq n_k$ . For fixed basis this is the standard elliptical-potential lemma [Abbasi-Yadkori et al., 2011, Lem. 11] (resp. its sliding-window form when  $W < n_k$ ); the moving-basis case reduces to controlling the cumulative log-determinant change under basis updates, which follows from prefix concentration (Lem. B.17) via a routine Lipschitz bound.*

**Lemma B.3** (Windowed projected self-normalized concentration). *For a fixed  $a^* \in \mathbb{R}^r$  and  $\sigma_\varepsilon$ -sub-Gaussian noise on the projected covariates  $z_t = \widehat{U}_t^\top x_t \in \mathbb{R}^r$ , the windowed ridge estimator  $\widehat{a}_t = \widetilde{V}_t^{-1} \widetilde{b}_t$  satisfies, with probability at least  $1 - \delta/K$  per segment  $k$  and simultaneously for all  $t \in E_k$ ,*

$$\|\widehat{a}_t - a^*\|_{\widetilde{V}_t} \leq \beta_t^{(r,W)}, \quad \beta_t^{(r,W)} := \sigma_\varepsilon \sqrt{r \log(1 + WR_{\mathcal{A}}^2/(\lambda r)) + 2 \log(2K/\delta)} + \sqrt{\lambda} S_w.$$

*This is the windowed projected specialization of Abbasi-Yadkori et al. [2011, Thm. 1, 2].*

## B.2 Concentration of the lifted probe moment (load-bearing step)

This is the non-standard piece of the analysis; we derive it in full. We work with scaled-sphere probes  $u_t = \sqrt{d} v_t$ ,  $v_t \sim \text{Unif}(\mathbb{S}^{d-1})$ , which satisfy  $\|u_t\|_2 = \sqrt{d}$  *deterministically* and  $\mathbb{E}[u_t u_t^\top] = I_d$ . The probe moment operator  $\mathcal{K} : M \mapsto \mathbb{E}[(u^\top M u) u u^\top]$  is a linear bijection on the space of symmetric  $d \times d$  matrices.

**Lemma B.4** (Closed-form inverse of  $\mathcal{K}$  on symmetric matrices). *For scaled-sphere probes  $u = \sqrt{d} v$ ,  $v \sim \text{Unif}(\mathbb{S}^{d-1})$ ,  $\mathcal{K}$  acts on symmetric  $d \times d$  matrices as  $\mathcal{K}(M) = \frac{d}{d+2} (\text{tr}(M) I_d + 2M)$  with inverse  $\mathcal{K}^{-1}(N) = \frac{d+2}{2d} N - \frac{\text{tr}(N)}{2d} I_d$ . Moreover  $\|\mathcal{K}^{-1}\|_{\text{op} \rightarrow \text{op}} \leq 1$  on symmetric inputs.*

*Proof.* Use the fourth-moment identity for the uniform sphere  $\mathbb{E}[v_i v_j v_k v_l] = (\delta_{ij} \delta_{kl} + \delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) / (d(d+2))$  and the rescaling  $u = \sqrt{d} v$ :  $\mathbb{E}[(u^\top M u) u_k u_l] = d^2 \mathbb{E}[(v^\top M v) v_k v_l] = d^2 \cdot (2M_{kl} + \text{tr}(M) \delta_{kl}) / (d(d+2)) = \frac{d}{d+2} (2M_{kl} + \text{tr}(M) \delta_{kl})$  (using  $M$  symmetric). So  $\mathcal{K}(M) = \frac{d}{d+2} (\text{tr}(M) I_d + 2M)$ . Taking traces of  $N = \mathcal{K}(M)$  gives  $\text{tr}(N) = d \text{tr}(M)$ , so  $\text{tr}(M) = \text{tr}(N)/d$  and  $M = \frac{d+2}{2d} N - \frac{\text{tr}(N)}{2d} I_d$ .

For the operator-norm bound:  $\mathcal{K}^{-1}(N)$  is diagonalized in the eigenbasis of  $N$ ; if  $N$  has eigenvalues  $\mu_1, \dots, \mu_d$ , then  $\mathcal{K}^{-1}(N)$  has eigenvalues  $\nu_i = \frac{d+2}{2d} \mu_i - \frac{\text{tr}(N)}{2d}$ . Maximizing  $|\nu_i|$  over  $\|N\|_{\text{op}} = 1$  (i.e.  $\max_i |\mu_i| = 1$ ): at  $\mu_1 = 1, \mu_2 = \dots = \mu_d = -1$ ,  $\text{tr}(N) = 2 - d$  and  $\nu_1 = \frac{d+2-(2-d)}{2d} = 1$ ; at  $\mu_1 = -1, \mu_2 = \dots = \mu_d = 1$ ,  $\nu_1 = -1$ . All other eigenvalue patterns give  $|\nu_i| \leq 1$  by direct case analysis. So  $\|\mathcal{K}^{-1}\|_{\text{op} \rightarrow \text{op}} = 1$  on symmetric matrices for  $d \geq 2$ .  $\square$

**Lemma B.5** (Quadratic measurement identity). *On every probe round  $t \in \mathcal{T}_{\text{probe}}$ , with  $\delta_\sigma := \widehat{\sigma}^2 - \sigma_\varepsilon^2$  and  $m_t := \mathbb{E}[\varepsilon_t \theta_t \mid \mathcal{H}_{t-1}, u_t]$ ,*

$$\mathbb{E}[s_t \mid \sigma(\mathcal{H}_{t-1}, u_t)] = u_t^\top \widetilde{M}_t u_t + 2u_t^\top m_t - \delta_\sigma, \quad \|m_t\| \leq \epsilon_\times.$$

*Under exact probe conditions ( $\delta_\sigma = \epsilon_\times = 0$ ) this collapses to  $\mathbb{E}[s_t \mid \cdot] = u_t^\top \widetilde{M}_t u_t$ .*

*Proof.* Expand  $y_t^2 = (u_t^\top \theta_t)^2 + 2\varepsilon_t u_t^\top \theta_t + \varepsilon_t^2$  and condition on  $\sigma(\mathcal{H}_{t-1}, u_t)$ ; the middle term contributes  $2u_t^\top m_t$  with  $m_t$  as defined in the lemma, bounded by  $\epsilon_\times$  via the bounded state-noise coupling assumption (§2); the last term contributes  $\sigma_\varepsilon^2$ , and subtracting  $\widehat{\sigma}^2$  leaves  $-\delta_\sigma$ . Independence of  $u_t$  from  $(\theta_t, \mathcal{H}_{t-1})$  gives  $\mathbb{E}[(u_t^\top \theta_t)^2 \mid \mathcal{H}_{t-1}, u_t] = u_t^\top \widetilde{M}_t u_t$ .  $\square$

**Lemma B.6** (Lifted probe sample is near-unbiased). *Let  $G_t := \mathcal{K}^{-1}(s_t u_t u_t^\top)$  for scaled-sphere  $u_t = \sqrt{d} v_t$ ,  $v_t \sim \text{Unif}(\mathbb{S}^{d-1})$ . Then  $\mathbb{E}[G_t \mid \mathcal{H}_{t-1}] = \widetilde{M}_t + \widetilde{B}_t$  with bias*

$$\widetilde{B}_t = -\frac{\delta_\sigma}{d} I_d, \quad \|\widetilde{B}_t\|_{\text{op}} = \frac{|\delta_\sigma|}{d}.$$

*The bias is a scaled identity, hence shifts every eigenvalue uniformly and does not rotate eigenvectors of  $\widetilde{M}_t$ .*

*Proof.* By linearity of  $\mathcal{K}^{-1}$  and Lemma B.5,  $\mathbb{E}[G_t \mid \mathcal{H}_{t-1}] = \mathcal{K}^{-1}(\mathcal{K}(\widetilde{M}_t)) + 2\mathcal{K}^{-1}(\mathbb{E}[(u_t^\top m_t) u_t u_t^\top \mid \mathcal{H}_{t-1}]) - \delta_\sigma \mathcal{K}^{-1}(\mathbb{E}[u_t u_t^\top]) = \widetilde{M}_t + \widetilde{B}_t$ , with  $\widetilde{M}_t = \mathcal{K}^{-1}\mathcal{K}(\widetilde{M}_t)$ .

*Variance-bias term.*  $\mathbb{E}[u_t u_t^\top] = I_d$  exactly for scaled-sphere probes, so  $-\delta_\sigma \mathcal{K}^{-1}(I_d) = -\delta_\sigma [\frac{d+2}{2d} I_d - \frac{d}{2d} I_d] = -\delta_\sigma \cdot \frac{2}{2d} I_d = -\frac{\delta_\sigma}{d} I_d$ . This is a scaled identity, hence does not rotate eigenvectors of  $\widetilde{M}_t$ .

*State-noise coupling term.* The remaining contribution is  $2\mathcal{K}^{-1}(\mathbb{E}[(u_t^\top m_t) u_t u_t^\top \mid \mathcal{H}_{t-1}])$ . The  $(a, b)$ -entry of the inner expectation is  $\sum_c (m_t)_c \mathbb{E}[(u_t)_a (u_t)_b (u_t)_c]$ , a degree-three polynomial in  $u_t$ . For sphere-symmetric probes  $u_t \mapsto -u_t$  leaves the distribution invariant, so every odd-degree moment vanishes and  $\mathbb{E}[(u_t^\top m_t) u_t u_t^\top \mid \mathcal{H}_{t-1}] = 0$  exactly. Hence the coupling term contributes zero, and  $\widetilde{B}_t = -\frac{\delta_\sigma}{d} I_d$ .  $\square$

**Lemma B.7** (Uniform a.s. bound on  $G_t$ ). *For scaled-sphere probes  $\|u_t\| = \sqrt{d}$  deterministically; on the noise-truncation event  $|\varepsilon_t| \leq L_\varepsilon := \sigma_\varepsilon \sqrt{2 \log(2T/\delta)}$  (which holds with probability  $\geq 1 - \delta$  via union over  $T$  rounds),  $|s_t| \leq R_s := (\sqrt{d} S_w + L_\varepsilon)^2 + \widehat{\sigma}^2$ , and the centered lifted sample satisfies  $\|\tilde{X}_t\|_{\text{op}} \leq 2R_X$  with  $R_X := d \cdot R_s + S_w^2$ . So both the envelope and predictable variance are at most  $2R_X$  and  $R_X^2$  respectively.*

*Proof.* For scaled-sphere  $u_t$ ,  $\|u_t\| = \sqrt{d}$  exactly.  $|s_t| = |(u_t^\top \theta_t + \varepsilon_t)^2 - \widehat{\sigma}^2| \leq (\sqrt{d} S_w + L_\varepsilon)^2 + \widehat{\sigma}^2 = R_s$  using  $|u_t^\top \theta_t| \leq \|u_t\| \|\theta_t\| \leq \sqrt{d} S_w$  on  $\|\theta_t\| \leq S_w$  (orthonormal  $B_k^*$ ,  $\|w_t\| \leq S_w$ ). Then  $\|G_t\|_{\text{op}} = \|\mathcal{K}^{-1}(s_t u_t u_t^\top)\|_{\text{op}} \leq |s_t| \|u_t u_t^\top\|_{\text{op}} \cdot 1 = R_s \cdot d$  using  $\|\mathcal{K}^{-1}\|_{\text{op} \rightarrow \text{op}} \leq 1$  and  $\|u_t u_t^\top\|_{\text{op}} = d$ . Adding the deterministic  $S_w^2 = \|\widetilde{M}_t\|_{\text{op}}$  bound for the centering gives  $\|\tilde{X}_t\|_{\text{op}} \leq 2R_X$ .  $\square$

**Theorem B.8** (Matrix Bernstein for  $\widehat{M}_k$ ). *Let  $\widehat{M}_k := m_k^{-1} \sum_{t \in \mathcal{T}_k} G_t$  and  $X_t := G_t - \mathbb{E}[G_t \mid \mathcal{H}_{t-1}]$ . Conditional on Lemma B.7,  $\{X_t\}$  is a bounded matrix-valued MDS with  $\|X_t\|_{\text{op}} \leq 2R_X$  a.s. and  $\|\sum_{t \in \mathcal{T}_k} \mathbb{E}[X_t^2 \mid \mathcal{H}_{t-1}]\|_{\text{op}} \leq m_k R_X^2$ . Here  $\widetilde{B}$  denotes the segment-level bias; for scaled-sphere probes under the standing assumptions of §2,  $\widetilde{B}_t$  from Lem. B.6 collapses to the deterministic constant  $-(\delta_\sigma/d)I_d$ , so we drop the subscript when no ambiguity arises and write  $\widetilde{B}$  for the common value. Freedman's matrix inequality [Tropp, 2011] gives, with probability at least  $1 - \delta$ ,*

$$\|\widehat{M}_k - \bar{M}_k^{\text{probe}} - \widetilde{B}\|_{\text{op}} \leq 2R_X \sqrt{\log(2d/\delta)/m_k} + \frac{2R_X \log(2d/\delta)}{3m_k}. \quad (6)$$

*Proof.* The MDS property is Lemma B.6 (shifted by the segment-level bias  $\widetilde{B}$  from Lemma B.6, which is deterministic and thus enters as a constant offset). The a.s. bound  $\|X_t\|_{\text{op}} \leq 2R_X$  follows from  $\|G_t\|_{\text{op}} \leq R_X$  and  $\|\mathbb{E}[G_t]\|_{\text{op}} \leq R_X$ . The predictable variance bound follows from  $\mathbb{E}[X_t^2] \preceq \mathbb{E}[G_t^2] \preceq R_X^2 I$ . Applying the matrix Freedman inequality to the MDS  $\{X_t\}$  in  $\mathbb{R}_{\text{sym}}^{d \times d}$  with these two bounds yields (6); the leading  $\sqrt{\log(2d/\delta)/m_k}$  rate dominates the  $\log(2d/\delta)/m_k$  term for  $m_k \gtrsim \log(d/\delta)$ .  $\square$

**Proposition B.9** (Segment factorization).  $\bar{M}_k^{\text{probe}} = B_k^* \bar{S}_k^{\text{probe}} (B_k^*)^\top$  where  $\bar{S}_k^{\text{probe}} := m_k^{-1} \sum_{t \in \mathcal{T}_k} \mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}]$ . In particular,  $\text{range}(\bar{M}_k^{\text{probe}}) = \text{range}(B_k^*)$ .

*Proof.*  $\widetilde{M}_t = B_k^* \mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}] (B_k^*)^\top$  for  $t \in \mathcal{I}_k$ ; average over  $\mathcal{T}_k$  and pull  $B_k^*$  out. Range equality uses  $\lambda_r(\bar{S}_k^{\text{probe}}) \geq \lambda_{\min} > 0$  (Lem. B.1).  $\square$

565 **Corollary B.10** (Projector concentration). *If  $m_k$  is large enough that  $\|\widehat{M}_k - \bar{M}_k^{\text{probe}}\|_{\text{op}} \leq$*   
 566  *$\lambda_{\min}/2$ , then with probability  $\geq 1 - \delta$ ,*

$$\varepsilon_k := \|\widehat{P}_k - P_k^*\|_{\text{op}} \leq C_{\text{sub}} \sqrt{\log(2d/\delta)/m_k} + \Delta_\sigma, \quad C_{\text{sub}} := \frac{8R_X}{\lambda_{\min}}, \quad \Delta_\sigma := \frac{4|\delta_\sigma|/d}{\lambda_{\min}}. \quad (7)$$

567 *For scaled-sphere probes, the variance-misspecification component  $|\delta_\sigma|/d$  corresponds to a*  
 568 *scaled-identity bias (Lem. B.6) that is parallel to  $I_d$  and therefore preserves the eigenvectors*  
 569 *of  $\bar{M}_k^{\text{probe}}$ ; applying Davis–Kahan with the biased mean as reference removes the  $|\delta_\sigma|/d$*   
 570 *contribution to  $\Delta_\sigma$  at population level (Cor. 4.2).*

571 *Proof.* By Lemma B.6 (scaled-sphere),  $\mathbb{E}[\widehat{M}_k \mid \mathcal{H}] = \bar{M}_k^{\text{probe}} + \widetilde{B}$  with  $\widetilde{B} = -(\delta_\sigma/d)I_d$  (a  
 572 scaled identity), so  $\|\widetilde{B}\|_{\text{op}} = |\delta_\sigma|/d =: b_\sigma$ . Combining with Theorem B.8,  $\|\widehat{M}_k - \bar{M}_k^{\text{probe}}\|_{\text{op}} \leq$   
 573  $2R_X \sqrt{\log(2d/\delta)/m_k} + b_\sigma$ .  $\bar{M}_k^{\text{probe}}$  has rank exactly  $r$  with  $r$ -th eigenvalue  $\geq \lambda_{\min}$  (Lem. B.1  
 574 via Prop. B.9); Davis–Kahan [Bhatia, 2013, Thm. VII.3.1] gives  $\|\widehat{P}_k - P_k^*\|_{\text{op}} \leq 4\|\widehat{M}_k -$   
 575  $\bar{M}_k^{\text{probe}}\|_{\text{op}}/\lambda_{\min}$ , yielding (7).  $\square$

576 **Lemma B.11** (Variance estimator from probe residuals). *Split the first  $N$  probes of segment*  
 577  *$k$  in half: form  $\widehat{M}_k^{(N/2)}$  from the first  $N/2$  probes (matrix-Bernstein estimate), and define*  
 578 *the residual estimator*

$$\widehat{\sigma}^2 := \text{med}_{t \in \mathcal{T}_k, N/2 < t \leq N} \{ y_t^2 - u_t^\top \widehat{M}_k^{(N/2)} u_t \}.$$

579 *On the event of Theorem B.8 for  $\widehat{M}_k^{(N/2)}$ , with probability at least  $1 - \delta$ ,*

$$|\widehat{\sigma}^2 - \sigma_\varepsilon^2| \leq C \sigma_\varepsilon^2 \sqrt{\log(1/\delta)/N}.$$

580 *Proof sketch.* By Lemma B.5, on probe round  $t$ ,  $\mathbb{E}[y_t^2 - u_t^\top \widetilde{M}_t u_t \mid \mathcal{H}_{t-1}, u_t] = \sigma_\varepsilon^2 + O(\epsilon_\times)$ .  
 581 Substituting  $\widehat{M}_k^{(N/2)}$  for  $\widetilde{M}_t$  contributes  $|u_t^\top (\widehat{M}_k^{(N/2)} - \widetilde{M}_t) u_t| = O(\sqrt{1/N})$  on the matrix-  
 582 Bernstein event. The squared noise  $\varepsilon_t^2$  is sub-exponential with mean  $\sigma_\varepsilon^2$ ; the median of  
 583  $N/2$  i.i.d. samples concentrates at rate  $\sigma_\varepsilon^2 \sqrt{\log(1/\delta)/N}$  by standard concentration of the  
 584 empirical median.  $\square$

585 This chain (Lemmas B.4–B.7, Theorem B.8, Corollary B.10, Proposition B.9) gives a self-  
 586 contained subspace-recovery guarantee for SPSC.

### 587 B.3 Necessity of the probe-side conditions

588 The probe-side conditions of §2 (known probe distribution  $Q$  with full-dimensional coverage,  
 589 known noise variance, and bounded state-noise coupling  $\|\mathbb{E}[\varepsilon_t \theta_t]\| \leq \epsilon_\times$ ) are each individu-  
 590 ally necessary for identifiability of the predictable second moment  $\widetilde{M}_t$  from single-play scalar  
 591 rewards. The three propositions below construct two distinct conditional second moments  
 592 generating identical observation laws when the named condition is dropped.

593 *Scope.* These propositions establish identifiability obstructions in the unrestricted-second-  
 594 moment problem and serve as necessity checks for the probe-side modeling assumptions.  
 595 They are not lower bounds inside the rank- $r$  LDS class of §2: the counterexample second  
 596 moments need not be rank- $r$  or compatible with a stable LDS  $w_t$ .

597 **Proposition B.12** (Non-identification without known noise variance). *Fix  $t$ . Suppose*  
 598  *$\epsilon_\times = 0$  (exact state-noise orthogonality) and full probe coverage hold, but the conditional*  
 599 *noise variance  $v_t(u) := \mathbb{E}[\varepsilon_t^2 \mid \mathcal{H}_{t-1}, u]$  is completely unknown. Then  $\widetilde{M}_t$  is not identifiable*  
 600 *from the observable  $g_t(u) := \mathbb{E}[y_t^2 \mid \mathcal{H}_{t-1}, u]$ .*

601 *Proof.*  $g_t(u) = u^\top \widetilde{M}_t u + v_t(u)$ . For any small  $\delta \in (0, \sigma_\varepsilon^2/L^2]$ , set  $H := \delta I_d$  and define  
 602  $\widetilde{M}'_t := \widetilde{M}_t + H$  and  $v'_t(u) := v_t(u) - u^\top H u = v_t(u) - \delta \|u\|^2$ . On the truncation support  
 603  $\|u\| \leq L$ ,  $v'_t(u) \geq \sigma_\varepsilon^2 - \delta L^2 \geq 0$ , so  $v'_t$  is a valid conditional variance. The two pairs  $(\widetilde{M}_t, v_t)$   
 604 and  $(\widetilde{M}'_t, v'_t)$  produce identical observables  $g_t$ , so  $\widetilde{M}_t \neq \widetilde{M}'_t$  are indistinguishable.  $\square$

**Proposition B.13** (Non-identification without bounded state-noise coupling). *Fix  $t$ . Suppose  $\sigma_\varepsilon^2$  is known and probe coverage is full, but the cross-moment  $m_t(u) := \mathbb{E}[\theta_t \varepsilon_t \mid \mathcal{H}_{t-1}, u]$  is arbitrary (no bound  $\epsilon_\times$ ). Then  $\widetilde{M}_t$  is not identifiable from  $h_t(u) := \mathbb{E}[y_t^2 - \sigma_\varepsilon^2 \mid \mathcal{H}_{t-1}, u]$ .*

*Proof.*  $h_t(u) = u^\top \widetilde{M}_t u + 2u^\top m_t(u)$ . For any nonzero symmetric  $H$ , let  $\widetilde{M}'_t := \widetilde{M}_t + H$  and  $m'_t(u) := m_t(u) - \frac{1}{2}Hu$ . Then  $u^\top \widetilde{M}'_t u + 2u^\top m'_t(u) = u^\top \widetilde{M}_t u + u^\top H u + 2u^\top m_t(u) - u^\top H u = h_t(u)$  for every  $u$ , so the two pairs are observationally equivalent.  $\square$

**Proposition B.14** (Non-identification without full-dimensional coverage). *Suppose probe actions  $u_t$  lie in a proper subspace  $S \subsetneq \mathbb{R}^d$  almost surely. Then  $\widetilde{M}_t$  is not identifiable from either linear rewards  $y_t$  or quadratic observations  $y_t^2$ .*

*Proof.* Pick any nonzero symmetric  $H$  supported on  $S^\perp$  (i.e.  $H = P_{S^\perp} H P_{S^\perp} \neq 0$ , possible since  $S^\perp \neq \{0\}$ ). For every  $u \in S$ ,  $Hu \in S^\perp$  and  $u^\top Hu = 0$ , so  $u^\top (\widetilde{M}_t + H)u = u^\top \widetilde{M}_t u$  and  $u^\top (\theta_t + Hz) = u^\top \theta_t$  for any  $z \in S^\perp$ . Thus the pairs  $(\theta_t, \widetilde{M}_t)$  and  $(\theta_t + Hz, \widetilde{M}_t + H)$  generate identical observations on all probe rounds, so  $\widetilde{M}_t$  is not identifiable.  $\square$

**Remark B.15** (Joint necessity). *No two of the three conditions jointly suffice: Prop. B.12 fails only on (i); Prop. B.13 only on (ii); Prop. B.14 only on (iii). Each counterexample satisfies the remaining two conditions, so the three form a minimal identifiability set for the single-play quadratic measurement model.*

#### B.4 Lemmas for the interleaved analysis

**Lemma B.16** (Projected nonstationary UCB with time-varying basis). *Fix a segment  $\mathcal{I}_k$ . For each exploit round  $t \in E_k$ , define  $U_t := \widehat{U}_{t-1}$ ,  $P_t := U_t U_t^\top$ ,  $z_t(x) := U_t^\top x$ ,  $a_t^* := U_t^\top \theta_t$ , and (as in Alg. 1)  $\widetilde{V}_t = \lambda I_r + \sum_{s \in \mathcal{W}_t} z_t(x_s) z_t(x_s)^\top$ ,  $\widehat{a}_t = \widetilde{V}_t^{-1} \sum_{s \in \mathcal{W}_t} z_t(x_s) y_s$ . Assume on segment  $\mathcal{I}_k$ : (i) the windowed projected self-normalized event of Lem. B.3 holds uniformly over  $E_k$ ; (ii) Assumption B.2 holds; (iii)  $\|\theta_t\| \leq S_w$ ,  $\|x\| \leq R_{\mathcal{A}}$  for  $x \in \mathcal{A}_t$ . Suppose Alg. 1 uses an optimism correction satisfying*

$$\gamma_t \geq C_\gamma S_w \left(1 + R_{\mathcal{A}} \sqrt{W/\lambda}\right) \varepsilon_{k,t-1} + C_\gamma R_{\mathcal{A}} V_{k,t}(W), \quad (8)$$

where  $L_W := \log(1 + WR_{\mathcal{A}}^2/(\lambda r))$  and  $V_{k,t}(W) := \sum_{s, s+1 \in \mathcal{I}_k, t-W \leq s < t} \|\theta_{s+1} - \theta_s\|_2$ . Then on the same event,

$$\sum_{t \in E_k} \left( \max_{x \in \mathcal{A}_t} x^\top \theta_t - x_t^\top \theta_t \right) \leq \widetilde{\mathcal{O}}(r\sqrt{n_k}) + C_{\text{tv}} R_{\mathcal{A}} S_w \left(1 + R_{\mathcal{A}} \sqrt{W/\lambda}\right) \sum_{t \in E_k} \varepsilon_{k,t-1} + CR_{\mathcal{A}} W V_k^{\text{in}}. \quad (9)$$

*Proof.* Fix  $t \in E_k$  and decompose, for  $x \in \mathcal{A}_t$ ,  $x^\top \theta_t = z_t(x)^\top a_t^* + \zeta_t(x)$  with  $\zeta_t(x) = x^\top (I - P_t) \theta_t$ . Since  $\theta_t = P_k^* \theta_t$  within segment  $k$  and  $\|(I - P_t) P_k^*\|_{\text{op}} = \|P_t - P_k^*\|_{\text{op}} = \varepsilon_{k,t-1}$  for two same-rank orthogonal projectors,

$$|\zeta_t(x)| \leq R_{\mathcal{A}} S_w \varepsilon_{k,t-1}. \quad (10)$$

Rewrite the windowed regression in basis  $U_t$ . For  $s \in \mathcal{W}_t$ ,  $y_s = z_t(x_s)^\top a_t^* + b_{s,t} + \varepsilon_s$  with bias  $b_{s,t} = x_s^\top (I - P_t) \theta_s + x_s^\top P_t (\theta_s - \theta_t)$ . The first term has  $|x_s^\top (I - P_t) \theta_s| \leq R_{\mathcal{A}} S_w \varepsilon_{k,t-1}$  (again  $\theta_s = P_k^* \theta_s$ ). The second is at most  $R_{\mathcal{A}} \sum_{u=s}^{t-1} \|\theta_{u+1} - \theta_u\|_2$ .

By definition,  $\widehat{a}_t - a_t^* = \widetilde{V}_t^{-1} \sum_{s \in \mathcal{W}_t} z_t(x_s) \varepsilon_s + \widetilde{V}_t^{-1} \sum_{s \in \mathcal{W}_t} z_t(x_s) b_{s,t} - \lambda \widetilde{V}_t^{-1} a_t^*$ . On the self-normalized event of Lem. B.3 the noise+ridge contribution has weighted norm  $\leq \beta_t^{(r,W)} = \widetilde{\mathcal{O}}(\sqrt{r})$ .

For the deterministic subspace-bias part, let  $b_t^{\text{sub}} \in \mathbb{R}^{|\mathcal{W}_t|}$  have entries  $x_s^\top (I - P_t) \theta_s$ , with  $\|b_t^{\text{sub}}\|_\infty \leq R_{\mathcal{A}} S_w \varepsilon_{k,t-1}$ . Let  $Z_t$  be the matrix with rows  $z_t(x_s)^\top$ ,  $s \in \mathcal{W}_t$ . Since

642  $\sum_{s \in \mathcal{W}_t} z_t(x_s) z_t(x_s)^\top = Z_t^\top Z_t \preceq \tilde{V}_t$ ,  $\|\tilde{V}_t^{-1/2} Z_t^\top\|_{\text{op}} \leq 1$ , hence

$$\left\| \sum_{s \in \mathcal{W}_t} z_t(x_s) b_{s,t}^{\text{sub}} \right\|_{\tilde{V}_t^{-1}}^2 = \|\tilde{V}_t^{-1/2} Z_t^\top b_t^{\text{sub}}\|_2^2 \leq \|b_t^{\text{sub}}\|_2^2 \leq |\mathcal{W}_t| \|b_t^{\text{sub}}\|_\infty^2 \leq W R_{\mathcal{A}}^2 S_w^2 \varepsilon_{k,t-1}^2,$$

643 so the corresponding  $\tilde{V}_t^{-1}$ -norm is at most  $R_{\mathcal{A}} S_w \sqrt{W} \varepsilon_{k,t-1}$ . Therefore for any  $x$ ,  
 644  $|z_t(x)^\top \tilde{V}_t^{-1} \sum_s z_t(x_s) b_{s,t}^{\text{sub}}| \leq R_{\mathcal{A}} S_w \sqrt{W} \varepsilon_{k,t-1} w_t(x)$ , and since  $w_t(x) \leq \|x\|/\sqrt{\lambda}$ , this is  
 645 bounded by  $R_{\mathcal{A}} S_w \sqrt{W/\lambda} \varepsilon_{k,t-1} \|x\|$ , a quantity dominated by the  $\gamma_t \|x\|$  correction in (8).  
 646 The drift-bias part contributes  $\leq C R_{\mathcal{A}} W V_k^{\text{in}}$  in total by the standard windowed argument  
 647 (each increment appears in  $\leq W$  windows).

648 By the optimism index in Alg. 1, with  $x_t^* = \arg \max_{x \in \mathcal{A}_t} x^\top \theta_t$ , the instantaneous regret  
 649 obeys

$$\Delta_t \leq 2\beta_t^{(r,W)} w_t(x_t) + C_{\text{tv}} R_{\mathcal{A}} S_w \left(1 + R_{\mathcal{A}} \sqrt{W/\lambda}\right) \varepsilon_{k,t-1} + d_t,$$

650 where  $\sum_t d_t \leq C R_{\mathcal{A}} W V_k^{\text{in}}$ . Cauchy-Schwarz together with Assumption B.2 gives  
 651  $\sum_{t \in E_k} \beta_t^{(r,W)} w_t(x_t) \leq (\sup_{t \in E_k} \beta_t^{(r,W)}) \sqrt{n_k \cdot 2C_{\text{pot}} r L_W} = \tilde{\mathcal{O}}(r \sqrt{n_k})$  (using  $\beta_t^{(r,W)} =$   
 652  $\tilde{\mathcal{O}}(\sqrt{r})$ ). Summing yields (9).  $\square$

653 **Lemma B.17** (Uniform prefix subspace concentration). *Under the probe-side assumptions*  
 654 *of §2, scaled-sphere probes, exact centering  $\hat{\sigma}^2 = \sigma_\varepsilon^2$ , and  $\lambda_{\min} > 0$ , there exist constants*  
 655  *$C_{\text{sub}}, C_0 > 0$  such that, with probability at least  $1 - \delta/3$ , simultaneously for all segments  $k$*   
 656 *and all rounds  $t \in \mathcal{I}_k$ ,*

$$\varepsilon_{k,t} \leq 1 \text{ if } q_k(t) < q^*, \quad \varepsilon_{k,t} \leq C_{\text{sub}} \sqrt{\log(2KdT/\delta)/q_k(t)} \text{ if } q_k(t) \geq q^*,$$

657 where  $q^* := C_0 \log(2KdT/\delta)/\lambda_{\min}^2$ .

658 *Proof.* For a fixed segment  $k$  and probe-prefix size  $q$ , the lifted probe estimator  
 659  $q^{-1} \sum_{s \in \mathcal{T}_k: s \leq t} G_s$  satisfies a matrix Bernstein/Freedman bound (Thm. B.8) at deviation  
 660  $C \sqrt{\log(2KdT/\delta)/q}$  with probability  $\geq 1 - \delta/(3KT)$ . By Lem. B.1 the predictable probe-  
 661 time  $r$ th eigenvalue is  $\geq \lambda_{\min}$ . Whenever  $q \geq q^* = C_0 \log(2KdT/\delta)/\lambda_{\min}^2$  the pertur-  
 662 bation is at most a fixed fraction of the eigengap, and Davis-Kahan (Cor. B.10) gives  
 663  $\|\hat{P}_t - P_k^*\|_{\text{op}} \leq C_{\text{sub}} \sqrt{\log(2KdT/\delta)/q}$ . A union bound over all  $K$  segments and all at most  
 664  $T$  probe prefixes yields the simultaneous statement; for  $q < q^*$  we use the trivial bound  
 665  $\|\hat{P}_t - P_k^*\|_{\text{op}} \leq 1$ .  $\square$

666 **Lemma B.18** (Cumulative interleaved subspace error). *On the event of Lemma B.17, for*  
 667 *every segment  $k$ ,*

$$\sum_{t \in E_k} \varepsilon_{k,t-1} \leq C \sqrt{\log(2KdT/\delta)} \frac{\ell_k}{\sqrt{m_k}} + C \frac{q^* \ell_k}{m_k}. \quad (11)$$

668 *Proof.* Group exploit rounds by  $j = q_k(t-1)$ . By the balanced-schedule condition (B)  
 669 each group has at most  $a_0 \lceil \ell_k/m_k \rceil$  exploit rounds. For  $j < q^*$  use  $\varepsilon_{k,t-1} \leq 1$ , contributing  
 670  $C q^* \ell_k/m_k$ . For  $j \geq q^*$ , Lemma B.17 gives  $\varepsilon_{k,t-1} \leq C_{\text{sub}} \sqrt{\log(2KdT/\delta)/j}$ , so  $\sum_{j=q^*}^{m_k} j^{-1/2} \leq$   
 671  $2\sqrt{m_k}$  yields the first term.  $\square$

## 672 B.5 Proof of Theorem 4.1

673 Fix segment  $\mathcal{I}_k$ . We decompose the costed regret on  $\mathcal{I}_k$  into probe regret and exploitation  
 674 regret.

675 **Probe regret.** On a probe round  $t \in \mathcal{T}_k$ , Alg. 1 plays  $x_t = u_t \sim Q$  rather than an action  
 676 in  $\mathcal{A}_t$ . Since  $\|x\| \leq R_{\mathcal{A}}$  for  $x \in \mathcal{A}_t$ ,  $\|u_t\| \leq R_Q$ , and  $\|\theta_t\| \leq S_w$ ,  $\max_{x \in \mathcal{A}_t} x^\top \theta_t - u_t^\top \theta_t \leq$   
 677  $(R_{\mathcal{A}} + R_Q) S_w$ , plus probe cost  $c$ . So each probe contributes at most  $A := (R_{\mathcal{A}} + R_Q) S_w + c$ ,  
 678 giving  $\text{Reg}_{\text{probe}}(\mathcal{I}_k) \leq A m_k$ .

679 **Exploitation regret.** By Lemmas B.16 and B.18,

$$\text{Reg}_{\text{exploit}}(\mathcal{I}_k) \leq \tilde{\mathcal{O}}(r\sqrt{n_k}) + B \frac{\ell_k}{\sqrt{m_k}} + CR_{\mathcal{A}}S_w \left(1 + R_{\mathcal{A}}\sqrt{W/\lambda}\right) \frac{q^*\ell_k}{m_k} + CR_{\mathcal{A}}WV_k^{\text{in}},$$

680 absorbing constants and logs into  $B = CR_{\mathcal{A}}S_w(1 + R_{\mathcal{A}}\sqrt{W/\lambda})\sqrt{\log(2KdT/\delta)}$ .

681 **Per-segment bound.** Combining,

$$\text{DynReg}^{(c)}(\mathcal{I}_k) \leq \tilde{\mathcal{O}}(r\sqrt{n_k}) + Am_k + B \frac{\ell_k}{\sqrt{m_k}} + \tilde{\mathcal{O}}\left(\frac{q^*\ell_k}{m_k}\right) + CR_{\mathcal{A}}WV_k^{\text{in}}. \quad (12)$$

682 **Optimizing the probe-estimation tradeoff.** Ignoring the lower-order burn-in term,  
 683 minimize  $f_k(m) := Am + B\ell_k m^{-1/2}$ . Setting  $f'_k(m) = A - B\ell_k/(2m^{3/2}) = 0$  gives  $m_k^* =$   
 684  $(B\ell_k/(2A))^{2/3}$ , and  $Am_k^* + B\ell_k/\sqrt{m_k^*} \leq C A^{1/3} B^{2/3} \ell_k^{2/3}$ . At  $m_k^*$  the burn-in contribution is  
 685  $\tilde{\mathcal{O}}(\ell_k^{1/3}/\lambda_{\min}^2)$  (after absorbing  $A, B$  constants). If  $\ell_k < q^*$ , the segment is charged trivially  
 686 by  $\tilde{\mathcal{O}}(\ell_k)$ , absorbed into the same burn-in order. Hence

$$\text{DynReg}^{(c)}(\mathcal{I}_k) \leq \tilde{\mathcal{O}}(r\sqrt{n_k}) + \tilde{\mathcal{O}}(A^{1/3} B^{2/3} \ell_k^{2/3}) + \tilde{\mathcal{O}}(\ell_k^{1/3}/\lambda_{\min}^2) + CR_{\mathcal{A}}WV_k^{\text{in}}. \quad (13)$$

687 **Sum over segments.** By Cauchy-Schwarz,  $\sum_k r\sqrt{n_k} \leq r\sqrt{KT}$ . By Hölder,  $\sum_k \ell_k^{2/3} \leq$   
 688  $K^{1/3} T^{2/3}$  and  $\sum_k \ell_k^{1/3} \leq K^{2/3} T^{1/3}$ , and  $\sum_k V_k^{\text{in}} = V_{\text{in}}$ . Summing (13) over  $k$  gives

$$\text{DynReg}_T^{(c)} \leq \tilde{\mathcal{O}}(r\sqrt{KT}) + \tilde{\mathcal{O}}(A^{1/3} B^{2/3} K^{1/3} T^{2/3}) + \tilde{\mathcal{O}}(K^{2/3} T^{1/3}/\lambda_{\min}^2) + O(R_{\mathcal{A}}WV_{\text{in}}).$$

689 Union-bounding the windowed self-normalized event, the prefix subspace concentration, and  
 690 the noise event at total  $\delta$  yields (4). For fixed  $K$ , the leading term is  $\tilde{\mathcal{O}}(r\sqrt{T})$ , the probe  
 691 term is  $\tilde{\mathcal{O}}(T^{2/3})$ , and the burn-in term  $\tilde{\mathcal{O}}(T^{1/3})$  is lower order. When  $V_{\text{in}} = 0$ , the third  
 692 term vanishes and we recover Corollary 4.3.  $\square$

## 693 B.6 Proof of Corollary 4.4

694 By Theorem B.8, with probability  $\geq 1 - \delta$  every eigenvalue of  $\widehat{M}_k$  is within  $\tau_k^{\text{rank}} :=$   
 695  $2R_X\sqrt{\log(2d/\delta)/m_k}$  of the corresponding eigenvalue of  $\bar{M}_k^{\text{probe}} + \widehat{B}$  (Weyl's inequality ap-  
 696 plied to the matrix in (6)). The population spectrum has exactly  $r$  eigenvalues  $\geq \lambda_{\min}$  and  
 697 the remaining  $d - r$  are zero (Prop. B.9). Under the eigengap hypothesis  $\lambda_{\min} \geq 4\tau_k^{\text{rank}}$ ,  
 698 thresholding at  $2\tau_k^{\text{rank}}$  returns exactly  $r$  indices with probability  $\geq 1 - \delta$ . Condition on this  
 699 event; the regret analysis of Theorem 4.1 applies verbatim with the estimated rank equal to  
 700 the true  $r$ , so (4) holds with an extra  $\delta$  in the union bound.  $\square$

## 701 B.7 Proof of Proposition 4.5 (SPSC-Adaptive)

702 The detector statistic of Alg. 2 compares two non-overlapping rolling-window lifted-moment  
 703 estimators of length  $W_{\text{det}}$ ,  $\widehat{M}_t^{\text{recent}}$  and  $\widehat{M}_t^{\text{past}}$ , via  $S_t := \|\widehat{M}_t^{\text{recent}} - \widehat{M}_t^{\text{past}}\|_{\text{op}}$ , and triggers  
 704 a reset whenever  $S_t > 2\eta_{\text{det}}$ , where  $\eta_{\text{det}} := C_{\text{sub}}\sqrt{\log(dT/\delta_{\text{FA}})/W_{\text{det}}}$  is the calibration  
 705 radius. The proof has two parts: bounding the false-alarm probability under  $H_0$  (no change  
 706 inside the comparison window) and bounding the detection delay under  $H_1$  (a change of  
 707 size  $\Delta_k \geq 2b$  has occurred), and then propagating the resulting boundary error through  
 708 Theorem 4.1.

709 **(i) False-alarm control under  $H_0$ .** Fix a round  $t$  and condition on  $H_0$ : the predictable  
 710 second moment is constant across the past and recent windows that produced  $\widehat{M}_t^{\text{past}}$  and  
 711  $\widehat{M}_t^{\text{recent}}$ . Apply Theorem B.8 (matrix Bernstein for the lifted probe estimator) to each  
 712 window separately at confidence  $\delta_{\text{FA}}/(2T)$  to get

$$\|\widehat{M}_t^{\text{recent}} - \bar{M}_t^{\text{recent}}\|_{\text{op}} \leq \eta_{\text{det}}, \quad \|\widehat{M}_t^{\text{past}} - \bar{M}_t^{\text{past}}\|_{\text{op}} \leq \eta_{\text{det}},$$

each with probability at least  $1 - \delta_{\text{FA}}/(2T)$ , where  $\bar{M}_t^{\text{recent}}$  and  $\bar{M}_t^{\text{past}}$  are the predictable averages on the corresponding windows. Under  $H_0$  they coincide, so by the triangle inequality  $S_t \leq \|\bar{M}_t^{\text{recent}} - \bar{M}_t^{\text{recent}}\|_{\text{op}} + \|\bar{M}_t^{\text{past}} - \bar{M}_t^{\text{past}}\|_{\text{op}} \leq 2\eta_{\text{det}}$  with probability at least  $1 - \delta_{\text{FA}}/T$ , i.e.  $\Pr(S_t > 2\eta_{\text{det}} \mid H_0) \leq \delta_{\text{FA}}/T$ . A union bound over  $t \in [T]$  gives total false-alarm probability at most  $\delta_{\text{FA}}$ .

**(ii) Detection delay under  $H_1$ .** Let  $\tau_k$  be a true change point with eigengap  $\Delta_k := \|P_k^* - P_{k-1}^*\|_{\text{op}} \geq 2b$  for the threshold  $b > 2\eta_{\text{det}}$ . Take  $t = \tau_k + W_{\text{det}}$ : the recent window now lies entirely in segment  $k$  while the past window lies entirely in segment  $k-1$ . By the segment-factorization of the predictable probe moment (Proposition B.9) and probe-time excitation (Lemma B.1),  $\|\bar{M}_t^{\text{recent}} - \bar{M}_t^{\text{past}}\|_{\text{op}} \geq \lambda_{\min} \Delta_k \geq 2b\lambda_{\min}$ . By the same matrix-Bernstein concentration as in (i),  $S_t \geq \|\bar{M}_t^{\text{recent}} - \bar{M}_t^{\text{past}}\|_{\text{op}} - 2\eta_{\text{det}} \geq 2b\lambda_{\min} - 2\eta_{\text{det}} > 2\eta_{\text{det}}$  with probability at least  $1 - \delta_{\text{FA}}/T$  (using  $b > 2\eta_{\text{det}}$  after absorbing  $\lambda_{\min}$  into  $b$ ). Therefore the detector triggers no later than  $\hat{\tau}_k = \tau_k + W_{\text{det}}$ , giving detection delay  $D_k \leq W_{\text{det}}$ .

**(iii) Boundary-error propagation.** On the detector good event  $\mathcal{E}_{\text{det}} = \{\text{no false alarm and every change detected within delay } D_k \leq W_{\text{det}}\}$ , which holds with probability at least  $1 - \delta_{\text{FA}}$  by parts (i)-(ii), the estimated boundaries  $\hat{\tau}_k$  satisfy  $|\hat{\tau}_k - \tau_k| \leq W_{\text{det}}$ . Splitting the horizon at  $\hat{\tau}_k$  and applying Theorem 4.1 on each estimated segment, the only difference from the oracle-boundary analysis is the contribution of the rounds in  $[\tau_k, \hat{\tau}_k)$  where the algorithm is still using the old basis  $\hat{U}_{\tau_k-1}$  on data drawn from segment  $k$ . Each such round contributes at most  $2R_{\mathcal{A}}S_w$  to the instantaneous regret, so  $\sum_k 2R_{\mathcal{A}}S_w D_k \leq 2R_{\mathcal{A}}S_w K W_{\text{det}} = O(KW_{\text{det}})$ . Combining,

$$\text{DynReg}_T^{(c)}(\text{Alg. 2}) \leq \text{DynReg}_T^{(c)}(\text{Alg. 1}) + O(KW_{\text{det}})$$

on  $\mathcal{E}_{\text{det}}$ , which is the claim. The total failure probability is  $\delta + \delta_{\text{FA}}$ , absorbed into the overall  $\delta$  budget by re-tuning constants.  $\square$

## C Probe distributions: scaled sphere and Gaussian truncation

The analysis in §B.2 uses scaled-sphere probes  $u_t = \sqrt{d}v_t$ ,  $v_t \sim \text{Unif}(\mathbb{S}^{d-1})$ , which satisfy  $\|u_t\|_2 = \sqrt{d}$  exactly. No truncation is required: the matrix-Bernstein envelope  $R_X = dR_s + S_w^2$  holds deterministically given the noise-truncation event of Lemma B.7.

In our implementation, probes are generated via  $u \leftarrow \sqrt{d}\tilde{u}/\|\tilde{u}\|$  for  $\tilde{u} \sim \mathcal{N}(0, I_d)$ , which is exactly the scaled-sphere distribution.

For isotropic-Gaussian probes (no rescaling),  $u_t \sim \mathcal{N}(0, I_d)$ , the Laurent–Massart  $\chi^2$  tail [Laurent and Massart, 2000, Lem. 1] gives  $\Pr(\|u_t\| > L) \leq \delta/T$  at  $L = \sqrt{2d \log(2T/\delta)}$ ; union-bounding over  $|\mathcal{T}_{\text{probe}}| \leq T$  gives  $\Pr(\bigcap_t \{\|u_t\| \leq L\}) \geq 1 - \delta$ . On this event the results of §B.2 apply with  $\|u_t\| \leq L$  (replacing  $\sqrt{d}$ ), adding only a  $\log(T/\delta)$  factor to  $R_X$  and  $C_{\text{sub}}$  that is absorbed in  $\tilde{\mathcal{O}}(\cdot)$ .

## D Algorithms: pseudocode and adaptive variant

This appendix gives the full pseudocode for SPSC (oracle boundaries, Alg. 1) and SPSC-Adaptive (online change-point detection, Alg. 2), and explains the adaptive design choices (detector form, threshold, recovery phase, burn-in) that tie the tuning back to the guarantee of Proposition 4.5.

## 752 D.1 SPSC (oracle boundaries)

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### Algorithm 1 SPSC: Single-Play Subspace-Calibrated Optimism (oracle boundaries)

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1: Input: segments  $\{\mathcal{I}_k\}_{k=1}^K$ , probe schedule  $\mathcal{T}_{\text{probe}}$ ,  $\lambda > 0$ , probe distribution  $Q$ , rank  $r$ ,
   window  $W$ .
2: Initialize  $\hat{U}_0 \in \mathbb{R}^{d \times r}$  arbitrarily with orthonormal columns.
3: for  $k = 1, \dots, K$  do ▷ outer loop over segments
4:    $M_{\text{acc}} \leftarrow 0$ ;  $N_k \leftarrow 0$ ;  $\hat{U}_t \leftarrow \hat{U}_{t-1}$ . ▷ reset segment-local accumulator and probe
   counter
5:   for  $t \in \mathcal{I}_k$  do
6:     if  $t \in \mathcal{T}_{\text{probe}}$  then ▷ probe round
7:       draw  $u_t \sim Q$ ; play  $x_t = u_t$ ; observe  $y_t$ .
8:        $s_t \leftarrow y_t^2 - \hat{\sigma}^2$ ;  $G_t \leftarrow \mathcal{K}^{-1}(s_t u_t u_t^\top)$ .
9:        $M_{\text{acc}} \leftarrow M_{\text{acc}} + G_t$ ;  $N_k \leftarrow N_k + 1$ ;  $\hat{U}_t \leftarrow \text{TopEig}_r(M_{\text{acc}}/N_k)$ .
10:    else ▷ exploitation round
11:       $\mathcal{W}_t \leftarrow \{s \in \mathcal{I}_k \setminus \mathcal{T}_{\text{probe}} : t-W \leq s < t\}$ ;  $z_t(x) \leftarrow \hat{U}_{t-1}^\top x$ .
12:       $\tilde{V}_t \leftarrow \lambda I_r + \sum_{s \in \mathcal{W}_t} z_t(x_s) z_t(x_s)^\top$ ,  $\tilde{b}_t \leftarrow \sum_{s \in \mathcal{W}_t} z_t(x_s) y_s$ .
13:       $\hat{a}_t \leftarrow \tilde{V}_t^{-1} \tilde{b}_t$ ; choose  $x_t = \arg \max_{x \in \mathcal{A}_t} \{z_t(x)^\top \hat{a}_t + \beta_t^{(r, W)} \|z_t(x)\|_{\tilde{V}_t^{-1}} +$ 
         $\gamma_t \|x\|_2\}$ .
14:      observe  $y_t$ ;  $\hat{U}_t \leftarrow \hat{U}_{t-1}$ .
15:    end if
16:  end for
17: end for

```

---

## 753 D.2 SPSC-Adaptive (unknown boundaries)

754 SPSC-Adaptive recovers segment boundaries from probe data, removing the oracle assumption  
 755 of Alg. 1.

756 **Detector.** At each probe round the algorithm maintains two non-overlapping rolling-  
 757 window estimators of the lifted second moment:  $\hat{M}_t^{\text{recent}}$  over the last  $n$  probes and  $\hat{M}_t^{\text{past}}$   
 758 over the segment-accumulated probes preceding that window. The statistic

$$S_t := \|\hat{M}_t^{\text{recent}} - \hat{M}_t^{\text{past}}\|_{\text{op}} \quad (14)$$

759 is compared to a threshold  $b$ ; whenever  $S_t > b$  the algorithm declares a change and enters  
 760 a RECOVERY phase.

761 **Recovery phase.** Recovery deploys  $m_{\text{relearn}}$  consecutive probe rounds with no exploitation.  
 762 The block-probe design serves two purposes: (i) by Proposition B.9,  $m_{\text{relearn}}$   
 763 back-to-back probes produce a subspace estimate  $\hat{U}$  with projector error  $\|\hat{P} - P^\star\|_{\text{op}} =$   
 764  $O(1/\sqrt{m_{\text{relearn}}})$  on a  $1 - \delta$  event; (ii) by zeroing out stale buffers it prevents the windowed  
 765 ridge estimator from carrying information from the previous regime into the new one. After  
 766  $m_{\text{relearn}}$  rounds the algorithm transitions back to NORMAL probe-exploitation interleaving  
 767 with probe rate  $\mu$ .

768 **Threshold calibration and detection delay.** Under  $H_0$  (no change), matrix Freedman  
 769 applied to the two windows gives  $\Pr(S_t > 2\eta_{\text{det}}) \leq \delta_{\text{FA}}/T$  with  $\eta_{\text{det}} = C_{\text{sub}} \sqrt{\log(dT/\delta_{\text{FA}})/n}$ ;  
 770 choosing  $b = 2\eta_{\text{det}}$  and union-bounding gives at most  $\delta_{\text{FA}}$  false alarms across the horizon.  
 771 Under  $H_1$  at time  $\tau_k$ , a separation  $\Delta_k \geq 2b$  forces a fire within delay  $D_{\text{max}} = 2n/\mu + \tau_{\text{burn}}$ ,  
 772 where the  $2n/\mu$  term is the time for the recent window to refill with post-change probes, and  
 773  $\tau_{\text{burn}}$  is a short burn-in period after recovery that prevents the detector from firing on its  
 774 own freshly-reset state. The overall regret overhead is  $2R_{\mathcal{A}} S_w \sum_k D_k = O(KW_{\text{det}})$ , which  
 775 is lower-order in the rate of Proposition 4.5.

---

**Algorithm 2 SPSC-Adaptive:** unknown boundaries via CUSUM-style detection

---

```
1: Input: probe rate  $\mu \in (0, 1)$ , detection window  $n$ , threshold  $b$ , burn-in  $\tau_{\text{burn}}$ , relearning  
   budget  $m_{\text{relearn}}$ ,  $W$ ,  $\lambda$ .  
2: phase  $\leftarrow$  RECOVERY; rec  $\leftarrow$  0;  $\widehat{M}_{\text{acc}} \leftarrow 0$ .  
3: for  $t = 1, \dots, T$  do  
4:   if phase = RECOVERY then  $\triangleright$  block-probe to rebuild subspace  
5:     probe: draw  $u_t$ , play  $x_t = u_t$ , observe  $y_t$ ; update  $\widehat{M}_{\text{acc}}$ ; rec  $+= 1$ .  
6:     if rec =  $m_{\text{relearn}}$  then  
7:        $\widehat{U} \leftarrow \text{TopEig}_r(\widehat{M}_{\text{acc}}/m_{\text{relearn}})$ ; reset buffers; phase  $\leftarrow$  NORMAL.  
8:     end if  
9:   else  $\triangleright$  interleave probes at rate  $\mu$   
10:    if Bernoulli( $\mu$ ) probe round then  
11:      update lifted estimator  $\widehat{M}_{\text{acc}}$  and refresh rolling windows.  
12:      if  $S_t > b$  (Eq. 14) and detector is armed then  
13:        phase  $\leftarrow$  RECOVERY; rec  $\leftarrow$  0; reset.  
14:      end if  
15:    else  
16:      windowed projected ridge-UCB step as in Alg. 1 (exploit branch).  
17:    end if  
18:  end if  
19: end for
```

---

776 **Tuning used in experiments.** Across all experiments reported in §5 and the appendix  
777 we use  $\mu = 0.1$ ,  $n = W_{\text{det}} = 50$ ,  $\tau_{\text{burn}} = 100$ ,  $m_{\text{relearn}} = 30$ , and  $b$  set from the CUSUM  
778 threshold `cusum_threshold` = 3.0. These are held fixed across datasets and  $(d, r)$  cells; no  
779 per-dataset retuning is done. Sensitivity to the probe rate  $\mu$  and the detection window  $n$  is  
780 reported in §G.

## 781 E Experimental setup

782 **Synthetic phase-transition benchmark.**  $\theta_t = B_k^* w_t$  with piecewise-constant orthonormal  
783  $B_k^* \in \mathbb{R}^{d \times r}$ . Per segment, latent state evolves as  $w_t = A_k w_{t-1} + \eta_{t-1}$  with spectral radius  
784  $\rho(A_k) = 0.99$  and Gaussian innovation  $\eta_{t-1} \sim \mathcal{N}(0, 0.04^2 I_r)$ . Reward noise  $\varepsilon_t \sim \mathcal{N}(0, 0.09)$ ;  
785 40 actions per round drawn i.i.d. uniformly on the unit sphere.  $K=10$  segments of equal  
786 length over  $T=5,000$ . Probe distribution  $Q$  isotropic Gaussian, probe cost  $c=0.1$ , probe  
787 period 50.

788 **UCI datasets (Coverttype, Pendigits, Satimage).** Features are the raw attribute  
789 vectors  $\phi(x) \in \mathbb{R}^{d_0}$  reduced to ambient dimension  $d$  via Gaussian random projection  $R \in$   
790  $\mathbb{R}^{d \times d_0}$ . For Coverttype, segments correspond to disjoint size- $K$  cover-type subsets of the 7  
791 classes; for Pendigits and Satimage, to disjoint digit/class subsets,  $K=10$ . Actions at round  
792  $t$  are  $|\mathcal{A}_t| = 40$  examples drawn from the active segment’s class pool. Reward is centered  
793 target label.  $\lambda=0.01$ , probe period 10, window  $W=400$ ,  $\delta=0.05$ ,  $c=0.02$ .

794 **Pixelized MNIST / Fashion-MNIST.** Images flattened, pixel-normalized, then pro-  
795 jected to  $d$  via random projection. Segments are class-subset swaps as above. Same hyper-  
796 parameters as UCI datasets.

797 **MovieLens-100K with real ratings.** Standard MovieLens-100K user-item matrix cen-  
798 tered by per-item mean. Features  $\phi(\text{user}, \text{item})$  are item-side latent factors from a rank- $d$   
799 SVD of the observed-entry mask; reward is the raw user rating (1–5), centered. Segments  
800 correspond to disjoint time-bucketed user cohorts,  $K=10$ . This is a *fully real* benchmark  
801 (we do *not* generate rewards from a low-rank surrogate), so the data is only approximately  
802 low-rank and provides a stringent stress test.  $T=5,000$ , same hyperparameters.

**Warfarin clinical dosing.** IWPC cohort statistics ( $\sim 5,700$  patients);  $d=93$  features (demographics, VKORC1/CYP2C9 genotype, medications, interactions). Reward is negative absolute deviation from the therapeutic dose,  $K=8$  patient-cohort segments,  $T=5,000$ ,  $\lambda=1.0$ , probe period 10,  $W=200$ ,  $c=0.1$ , 10 seeds.

**Baseline hyperparameters.** All baselines inherit  $\lambda, \delta$  from SPSC when applicable. D-LinUCB uses discount factor 0.998; SW-LinUCB uses window  $W$ ; Restart-LinUCB restarts every  $T/K$  rounds (i.e. oracle boundaries); LinTS/SW-LinTS use posterior variance  $\sigma^2 = \delta^2$ . Oracle-LinUCB knows the true  $B_k^*$  and runs LinUCB in the projected  $\mathbb{R}^r$ .

**Subspace-aware baselines (LowOFUL, VOFUL): favorable adaptation.** Both LowOFUL and VOFUL were designed for a *fixed* low-rank representation, so running them in our piecewise setting requires a choice for the subspace they consume. To avoid conflating subspace acquisition with exploitation, we give these baselines the same refreshed projector  $\hat{U}$  that SPSC uses (recomputed every 20 rounds after a 30-round warm-up), so every subspace-aware method sees an identical low-rank representation at every round. This is a *favorable* adaptation, since it removes representation quality as a possible source of SPSC’s empirical gains, and any remaining gap is attributable to SPSC’s windowed head and drift handling rather than to better subspace estimation. Running LowOFUL/VOFUL with their own stationary subspace estimators (not shown) yields strictly worse baseline numbers and larger SPSC margins.

**Implementation and compute.** All experiments use NumPy/SciPy only. A full  $(d, r)$  grid (9 cells, 10 seeds, nine per-cell baselines plus Oracle-LinUCB and the two SPSC variants; BOSS and Jedra appear only in the §5.6 grid) takes 6–20 hours on a single 16-core CPU. Covertypes is the longest ( $T=10,000$ ).

## F Per-cell tables: every dataset, every cell

Tables 8–15 give the full per-cell results (mean  $\pm$  SE costed regret over 10 seeds) for every method on every  $(d, r)$  cell. Cells where at least one SPSC variant beats every non-oracle baseline are highlighted **green**; the best non-oracle baseline per cell is underlined.

Table 8: **Covertypes** per-cell results. Mean  $\pm$  SE costed regret over 10 seeds. Green = SPSC variant beats every non-oracle baseline; underline = best non-oracle.  $T=10,000$ ,  $K=4$ . *Reading note (applies to all per-cell tables in this appendix):* rank-insensitive methods (LinUCB, SW-LinUCB, D-LinUCB, Restart-LinUCB, LinTS, SW-LinTS) act in ambient space; their regret depends on  $d$  but not on  $r$ , hence bit-identical values across the rows of each  $d$ -block.

| $d$ | $r$ | Oracle         | SPSC          | SPSC-Adp      | LowOFUL        | VOFUL          | LR-Rew         | SW-Lin         | D-Lin         | Rst-Lin        | LinUCB         | LinTS          | SW-LinTS       |
|-----|-----|----------------|---------------|---------------|----------------|----------------|----------------|----------------|---------------|----------------|----------------|----------------|----------------|
| 5   | 1   | 2013 $\pm$ 138 | 2655 $\pm$ 85 | 2114 $\pm$ 44 | 3748 $\pm$ 162 | 3736 $\pm$ 164 | 2379 $\pm$ 136 | 2651 $\pm$ 110 | 2502 $\pm$ 98 | 2661 $\pm$ 124 | 2694 $\pm$ 126 | 3633 $\pm$ 159 | 3064 $\pm$ 113 |
|     | 55  | 1              | 718 $\pm$ 51  | 1296 $\pm$ 21 | 1262 $\pm$ 32  | 1429 $\pm$ 32  | 1436 $\pm$ 32  | 1366 $\pm$ 44  | 1331 $\pm$ 20 | 1346 $\pm$ 21  | 1329 $\pm$ 21  | 1336 $\pm$ 21  | 1462 $\pm$ 25  |
| 10  | 1   | 2768 $\pm$ 29  | 4307 $\pm$ 62 | 4161 $\pm$ 63 | 5067 $\pm$ 83  | 5167 $\pm$ 130 | 5122 $\pm$ 61  | 4403 $\pm$ 41  | 4599 $\pm$ 57 | 4348 $\pm$ 37  | 4465 $\pm$ 49  | 5591 $\pm$ 78  | 5486 $\pm$ 73  |
|     | 20  | 4004 $\pm$ 55  | 5769 $\pm$ 54 | 5786 $\pm$ 84 | 7120 $\pm$ 93  | 7001 $\pm$ 72  | 7538 $\pm$ 81  | 5703 $\pm$ 34  | 5971 $\pm$ 40 | 5669 $\pm$ 40  | 5800 $\pm$ 50  | 7825 $\pm$ 41  | 7581 $\pm$ 44  |
| 30  | 1   | 5202 $\pm$ 48  | 6961 $\pm$ 95 | 7051 $\pm$ 81 | 8890 $\pm$ 110 | 8770 $\pm$ 93  | 9527 $\pm$ 96  | 6653 $\pm$ 34  | 6928 $\pm$ 30 | 6594 $\pm$ 51  | 6743 $\pm$ 34  | 9615 $\pm$ 105 | 9345 $\pm$ 90  |
|     | 105 | 1              | 554 $\pm$ 30  | 1023 $\pm$ 29 | 1010 $\pm$ 21  | 1134 $\pm$ 41  | 1124 $\pm$ 44  | 1026 $\pm$ 28  | 1075 $\pm$ 32 | 1082 $\pm$ 33  | 1073 $\pm$ 31  | 1075 $\pm$ 33  | 1109 $\pm$ 35  |
| 10  | 1   | 2154 $\pm$ 27  | 3452 $\pm$ 45 | 3405 $\pm$ 40 | 3775 $\pm$ 34  | 3758 $\pm$ 48  | 3869 $\pm$ 36  | 3920 $\pm$ 36  | 3959 $\pm$ 36 | 3909 $\pm$ 37  | 3943 $\pm$ 35  | 4198 $\pm$ 40  | 4168 $\pm$ 37  |
|     | 20  | 3233 $\pm$ 36  | 4600 $\pm$ 62 | 4507 $\pm$ 46 | 5297 $\pm$ 67  | 5325 $\pm$ 68  | 5621 $\pm$ 43  | 5302 $\pm$ 40  | 5383 $\pm$ 32 | 5313 $\pm$ 30  | 5355 $\pm$ 35  | 5826 $\pm$ 42  | 5734 $\pm$ 42  |
| 30  | 1   | 4165 $\pm$ 23  | 5529 $\pm$ 71 | 5450 $\pm$ 67 | 6476 $\pm$ 33  | 6556 $\pm$ 51  | 6989 $\pm$ 54  | 6360 $\pm$ 38  | 6473 $\pm$ 33 | 6324 $\pm$ 42  | 6400 $\pm$ 37  | 7085 $\pm$ 48  | 7029 $\pm$ 46  |

Table 9: **Pendigits** per-cell results.  $T=5,000$ ,  $K=10$ , 10 seeds.

| $d$ | $r$ | Oracle         | SPSC           | SPSC-Adp       | LowOFUL        | VOFUL          | LR-Rew         | SW-Lin        | D-Lin         | Rst-Lin       | LinUCB        | LinTS         | SW-LinTS      |
|-----|-----|----------------|----------------|----------------|----------------|----------------|----------------|---------------|---------------|---------------|---------------|---------------|---------------|
| 5   | 1   | 39.0 $\pm$ 2.0 | 2914 $\pm$ 159 | 5480 $\pm$ 391 | 9536 $\pm$ 199 | 9623 $\pm$ 138 | 6606 $\pm$ 77  | 1090 $\pm$ 9  | 1267 $\pm$ 8  | 1022 $\pm$ 14 | 1073 $\pm$ 15 | 6629 $\pm$ 42 | 4630 $\pm$ 41 |
|     | 55  | 1              | 18.0 $\pm$ 1.0 | 4019 $\pm$ 79  | 4073 $\pm$ 200 | 5913 $\pm$ 46  | 5939 $\pm$ 64  | 4542 $\pm$ 48 | 3863 $\pm$ 11 | 4056 $\pm$ 9  | 3620 $\pm$ 7  | 3658 $\pm$ 9  | 4343 $\pm$ 16 |
| 10  | 1   | 951 $\pm$ 5    | 2774 $\pm$ 58  | 2682 $\pm$ 93  | 3703 $\pm$ 71  | 3655 $\pm$ 67  | 4246 $\pm$ 28  | 3863 $\pm$ 11 | 4056 $\pm$ 9  | 3620 $\pm$ 7  | 3658 $\pm$ 9  | 4343 $\pm$ 16 | 4648 $\pm$ 9  |
|     | 20  | 1816 $\pm$ 4   | 2803 $\pm$ 60  | 2490 $\pm$ 56  | 3689 $\pm$ 54  | 3854 $\pm$ 58  | 4271 $\pm$ 39  | 3863 $\pm$ 11 | 4056 $\pm$ 9  | 3620 $\pm$ 7  | 3658 $\pm$ 9  | 4343 $\pm$ 16 | 4648 $\pm$ 9  |
| 30  | 1   | 2534 $\pm$ 6   | 3051 $\pm$ 62  | 2776 $\pm$ 45  | 3827 $\pm$ 45  | 3818 $\pm$ 39  | 4329 $\pm$ 20  | 3863 $\pm$ 11 | 4056 $\pm$ 9  | 3620 $\pm$ 7  | 3658 $\pm$ 9  | 4343 $\pm$ 16 | 4648 $\pm$ 9  |
|     | 105 | 1              | 11.0 $\pm$ 0.0 | 3011 $\pm$ 88  | 3120 $\pm$ 134 | 4254 $\pm$ 114 | 4277 $\pm$ 107 | 3051 $\pm$ 89 | 3297 $\pm$ 6  | 3400 $\pm$ 6  | 3152 $\pm$ 6  | 3184 $\pm$ 5  | 3117 $\pm$ 10 |
| 10  | 1   | 666 $\pm$ 3    | 2066 $\pm$ 64  | 1693 $\pm$ 38  | 2927 $\pm$ 74  | 2816 $\pm$ 55  | 3027 $\pm$ 43  | 3297 $\pm$ 6  | 3400 $\pm$ 6  | 3152 $\pm$ 6  | 3184 $\pm$ 5  | 3117 $\pm$ 10 | 3557 $\pm$ 4  |
|     | 20  | 1242 $\pm$ 3   | 2067 $\pm$ 62  | 1571 $\pm$ 67  | 2840 $\pm$ 41  | 2835 $\pm$ 35  | 3123 $\pm$ 18  | 3297 $\pm$ 6  | 3400 $\pm$ 6  | 3152 $\pm$ 6  | 3184 $\pm$ 5  | 3117 $\pm$ 10 | 3557 $\pm$ 4  |
| 30  | 1   | 1745 $\pm$ 4   | 2247 $\pm$ 58  | 1722 $\pm$ 58  | 2826 $\pm$ 75  | 2839 $\pm$ 56  | 3283 $\pm$ 19  | 3297 $\pm$ 6  | 3400 $\pm$ 6  | 3152 $\pm$ 6  | 3184 $\pm$ 5  | 3117 $\pm$ 10 | 3557 $\pm$ 4  |

Table 10: **Satimage** per-cell results.  $T=5,000$ ,  $K=10$ , 10 seeds.

| $d$ | $r$ | Oracle         | SPSC          | SPSC-Adp       | LowOFUL        | VOFUL         | LR-Rew        | SW-Lin       | D-Lin        | Rst-Lin      | LinUCB        | LinTS         | SW-LinTS      |
|-----|-----|----------------|---------------|----------------|----------------|---------------|---------------|--------------|--------------|--------------|---------------|---------------|---------------|
| 5   | 1   | 35.0 $\pm$ 2.0 | 1480 $\pm$ 88 | 1513 $\pm$ 123 | 2739 $\pm$ 17  | 2704 $\pm$ 18 | 2660 $\pm$ 24 | 655 $\pm$ 7  | 778 $\pm$ 6  | 629 $\pm$ 6  | 680 $\pm$ 6   | 2397 $\pm$ 14 | 1675 $\pm$ 16 |
| 55  | 1   | 18.0 $\pm$ 1.0 | 2439 $\pm$ 81 | 2015 $\pm$ 75  | 2913 $\pm$ 11  | 2911 $\pm$ 8  | 3450 $\pm$ 52 | 3051 $\pm$ 6 | 3201 $\pm$ 9 | 2779 $\pm$ 3 | 2848 $\pm$ 9  | 3403 $\pm$ 11 | 3781 $\pm$ 10 |
|     | 10  | 597 $\pm$ 3    | 1343 $\pm$ 41 | 1073 $\pm$ 49  | 2294 $\pm$ 99  | 2288 $\pm$ 96 | 2689 $\pm$ 62 | 3051 $\pm$ 6 | 3201 $\pm$ 9 | 2779 $\pm$ 3 | 2848 $\pm$ 9  | 3403 $\pm$ 11 | 3781 $\pm$ 10 |
|     | 20  | 1218 $\pm$ 4   | 1506 $\pm$ 34 | 1201 $\pm$ 33  | 2455 $\pm$ 106 | 2479 $\pm$ 65 | 2879 $\pm$ 55 | 3051 $\pm$ 6 | 3201 $\pm$ 9 | 2779 $\pm$ 3 | 2848 $\pm$ 9  | 3403 $\pm$ 11 | 3781 $\pm$ 10 |
|     | 30  | 1725 $\pm$ 8   | 1900 $\pm$ 38 | 1635 $\pm$ 40  | 2641 $\pm$ 54  | 2772 $\pm$ 47 | 3085 $\pm$ 53 | 3051 $\pm$ 6 | 3201 $\pm$ 9 | 2779 $\pm$ 3 | 2848 $\pm$ 9  | 3403 $\pm$ 11 | 3781 $\pm$ 10 |
| 105 | 1   | 16.0 $\pm$ 1.0 | 2484 $\pm$ 59 | 2333 $\pm$ 108 | 3638 $\pm$ 7   | 3655 $\pm$ 12 | 3911 $\pm$ 47 | 3572 $\pm$ 8 | 3671 $\pm$ 9 | 3441 $\pm$ 6 | 3484 $\pm$ 10 | 3600 $\pm$ 9  | 3891 $\pm$ 7  |
|     | 10  | 538 $\pm$ 3    | 1377 $\pm$ 44 | 1112 $\pm$ 49  | 2431 $\pm$ 87  | 2361 $\pm$ 78 | 2703 $\pm$ 40 | 3572 $\pm$ 8 | 3671 $\pm$ 9 | 3441 $\pm$ 6 | 3484 $\pm$ 10 | 3600 $\pm$ 9  | 3891 $\pm$ 7  |
|     | 20  | 1098 $\pm$ 5   | 1454 $\pm$ 33 | 1182 $\pm$ 34  | 2229 $\pm$ 57  | 2346 $\pm$ 46 | 2697 $\pm$ 39 | 3572 $\pm$ 8 | 3671 $\pm$ 9 | 3441 $\pm$ 6 | 3484 $\pm$ 10 | 3600 $\pm$ 9  | 3891 $\pm$ 7  |
|     | 30  | 1569 $\pm$ 5   | 1769 $\pm$ 32 | 1506 $\pm$ 33  | 2525 $\pm$ 42  | 2587 $\pm$ 80 | 2722 $\pm$ 25 | 3572 $\pm$ 8 | 3671 $\pm$ 9 | 3441 $\pm$ 6 | 3484 $\pm$ 10 | 3600 $\pm$ 9  | 3891 $\pm$ 7  |

Table 11: **Fashion-MNIST** per-cell results.  $T=5,000$ ,  $K=10$ , 10 seeds.

| $d$ | $r$ | Oracle       | SPSC          | SPSC-Adp       | LowOFUL        | VOFUL          | LR-Rew        | SW-Lin       | D-Lin         | Rst-Lin      | LinUCB        | LinTS         | SW-LinTS      |
|-----|-----|--------------|---------------|----------------|----------------|----------------|---------------|--------------|---------------|--------------|---------------|---------------|---------------|
| 55  | 5   | 539 $\pm$ 6  | 3773 $\pm$ 91 | 3792 $\pm$ 58  | 4487 $\pm$ 103 | 4475 $\pm$ 118 | 4719 $\pm$ 93 | 4169 $\pm$ 9 | 4324 $\pm$ 11 | 3801 $\pm$ 6 | 3868 $\pm$ 12 | 5183 $\pm$ 11 | 5357 $\pm$ 15 |
|     | 10  | 945 $\pm$ 5  | 3406 $\pm$ 73 | 3377 $\pm$ 92  | 4425 $\pm$ 126 | 4471 $\pm$ 142 | 4740 $\pm$ 59 | 4169 $\pm$ 9 | 4324 $\pm$ 11 | 3801 $\pm$ 6 | 3868 $\pm$ 12 | 5183 $\pm$ 11 | 5357 $\pm$ 15 |
|     | 20  | 1676 $\pm$ 6 | 3450 $\pm$ 62 | 3162 $\pm$ 81  | 4677 $\pm$ 119 | 4708 $\pm$ 75  | 4924 $\pm$ 56 | 4169 $\pm$ 9 | 4324 $\pm$ 11 | 3801 $\pm$ 6 | 3868 $\pm$ 12 | 5183 $\pm$ 11 | 5357 $\pm$ 15 |
| 105 | 5   | 399 $\pm$ 3  | 3200 $\pm$ 88 | 3208 $\pm$ 60  | 3365 $\pm$ 71  | 3522 $\pm$ 92  | 3688 $\pm$ 81 | 4152 $\pm$ 8 | 4236 $\pm$ 7  | 4016 $\pm$ 8 | 4034 $\pm$ 10 | 4376 $\pm$ 9  | 4496 $\pm$ 11 |
|     | 10  | 757 $\pm$ 6  | 2927 $\pm$ 76 | 2623 $\pm$ 84  | 3331 $\pm$ 42  | 3478 $\pm$ 51  | 3645 $\pm$ 65 | 4152 $\pm$ 8 | 4236 $\pm$ 7  | 4016 $\pm$ 8 | 4034 $\pm$ 10 | 4376 $\pm$ 9  | 4496 $\pm$ 11 |
|     | 20  | 1439 $\pm$ 4 | 2855 $\pm$ 75 | 2528 $\pm$ 59  | 3559 $\pm$ 80  | 3564 $\pm$ 45  | 3797 $\pm$ 38 | 4152 $\pm$ 8 | 4236 $\pm$ 7  | 4016 $\pm$ 8 | 4034 $\pm$ 10 | 4376 $\pm$ 9  | 4496 $\pm$ 11 |
| 200 | 5   | 393 $\pm$ 3  | 2949 $\pm$ 52 | 2877 $\pm$ 89  | 3134 $\pm$ 72  | 2869 $\pm$ 58  | 2948 $\pm$ 47 | 4099 $\pm$ 7 | 4158 $\pm$ 8  | 4059 $\pm$ 5 | 4059 $\pm$ 11 | 4101 $\pm$ 11 | 4206 $\pm$ 8  |
|     | 10  | 780 $\pm$ 4  | 2703 $\pm$ 52 | 2732 $\pm$ 101 | 2929 $\pm$ 43  | 2731 $\pm$ 63  | 2967 $\pm$ 26 | 4099 $\pm$ 7 | 4158 $\pm$ 8  | 4059 $\pm$ 5 | 4059 $\pm$ 11 | 4101 $\pm$ 11 | 4206 $\pm$ 8  |
|     | 20  | 1290 $\pm$ 5 | 2654 $\pm$ 52 | 2582 $\pm$ 67  | 3215 $\pm$ 56  | 3196 $\pm$ 61  | 3341 $\pm$ 44 | 4099 $\pm$ 7 | 4158 $\pm$ 8  | 4059 $\pm$ 5 | 4059 $\pm$ 11 | 4101 $\pm$ 11 | 4206 $\pm$ 8  |

Table 12: **MNIST** per-cell results.  $T=5,000$ ,  $K=10$ , 10 seeds.

| $d$ | $r$ | Oracle        | SPSC           | SPSC-Adp       | LowOFUL        | VOFUL          | LR-Rew         | SW-Lin        | D-Lin         | Rst-Lin       | LinUCB        | LinTS         | SW-LinTS      |
|-----|-----|---------------|----------------|----------------|----------------|----------------|----------------|---------------|---------------|---------------|---------------|---------------|---------------|
| 55  | 5   | 666 $\pm$ 5   | 5076 $\pm$ 88  | 5223 $\pm$ 109 | 6437 $\pm$ 152 | 6646 $\pm$ 62  | 6650 $\pm$ 116 | 5889 $\pm$ 16 | 6154 $\pm$ 12 | 5426 $\pm$ 10 | 5533 $\pm$ 9  | 6958 $\pm$ 13 | 7362 $\pm$ 20 |
|     | 10  | 1149 $\pm$ 9  | 4648 $\pm$ 77  | 4968 $\pm$ 93  | 6392 $\pm$ 114 | 6191 $\pm$ 118 | 6623 $\pm$ 77  | 5889 $\pm$ 16 | 6154 $\pm$ 12 | 5426 $\pm$ 10 | 5533 $\pm$ 9  | 6958 $\pm$ 13 | 7362 $\pm$ 20 |
|     | 20  | 2196 $\pm$ 11 | 4745 $\pm$ 76  | 4549 $\pm$ 82  | 6420 $\pm$ 63  | 6352 $\pm$ 120 | 6745 $\pm$ 58  | 5889 $\pm$ 16 | 6154 $\pm$ 12 | 5426 $\pm$ 10 | 5533 $\pm$ 9  | 6958 $\pm$ 13 | 7362 $\pm$ 20 |
| 105 | 5   | 582 $\pm$ 5   | 4658 $\pm$ 104 | 4844 $\pm$ 68  | 5184 $\pm$ 74  | 5305 $\pm$ 56  | 5807 $\pm$ 93  | 5992 $\pm$ 11 | 6073 $\pm$ 8  | 5749 $\pm$ 11 | 5811 $\pm$ 10 | 5892 $\pm$ 18 | 6222 $\pm$ 11 |
|     | 10  | 1029 $\pm$ 5  | 4298 $\pm$ 96  | 4229 $\pm$ 157 | 4867 $\pm$ 96  | 4917 $\pm$ 46  | 5555 $\pm$ 58  | 5992 $\pm$ 11 | 6073 $\pm$ 8  | 5749 $\pm$ 11 | 5811 $\pm$ 10 | 5892 $\pm$ 18 | 6222 $\pm$ 11 |
|     | 20  | 1869 $\pm$ 11 | 4273 $\pm$ 101 | 4076 $\pm$ 114 | 5020 $\pm$ 46  | 4918 $\pm$ 78  | 5651 $\pm$ 46  | 5992 $\pm$ 11 | 6073 $\pm$ 8  | 5749 $\pm$ 11 | 5811 $\pm$ 10 | 5892 $\pm$ 18 | 6222 $\pm$ 11 |
| 200 | 5   | 539 $\pm$ 2   | 4377 $\pm$ 84  | 4260 $\pm$ 81  | 4672 $\pm$ 62  | 4718 $\pm$ 113 | 4797 $\pm$ 52  | 5751 $\pm$ 10 | 5786 $\pm$ 6  | 5659 $\pm$ 11 | 5661 $\pm$ 8  | 5568 $\pm$ 10 | 5774 $\pm$ 7  |
|     | 10  | 907 $\pm$ 5   | 4114 $\pm$ 66  | 4023 $\pm$ 113 | 4184 $\pm$ 63  | 4375 $\pm$ 75  | 4818 $\pm$ 42  | 5751 $\pm$ 10 | 5786 $\pm$ 6  | 5659 $\pm$ 11 | 5661 $\pm$ 8  | 5568 $\pm$ 10 | 5774 $\pm$ 7  |
|     | 20  | 1675 $\pm$ 6  | 4052 $\pm$ 59  | 3828 $\pm$ 69  | 4379 $\pm$ 34  | 4413 $\pm$ 50  | 5022 $\pm$ 27  | 5751 $\pm$ 10 | 5786 $\pm$ 6  | 5659 $\pm$ 11 | 5661 $\pm$ 8  | 5568 $\pm$ 10 | 5774 $\pm$ 7  |

Table 13: **MovieLens** (fully real ratings) per-cell results.  $T=5,000$ ,  $K=10$ , 10 seeds.

| $d$ | $r$ | Oracle        | SPSC          | SPSC-Adp       | LowOFUL        | VOFUL          | LR-Rew        | SW-Lin       | D-Lin        | Rst-Lin       | LinUCB       | LinTS         | SW-LinTS      |
|-----|-----|---------------|---------------|----------------|----------------|----------------|---------------|--------------|--------------|---------------|--------------|---------------|---------------|
| 55  | 5   | 4013 $\pm$ 33 | 6213 $\pm$ 63 | 6060 $\pm$ 77  | 7676 $\pm$ 164 | 7504 $\pm$ 143 | 8103 $\pm$ 60 | 6313 $\pm$ 6 | 6363 $\pm$ 6 | 6239 $\pm$ 10 | 6258 $\pm$ 8 | 6396 $\pm$ 11 | 6587 $\pm$ 11 |
|     | 10  | 4802 $\pm$ 24 | 6250 $\pm$ 46 | 6015 $\pm$ 37  | 7689 $\pm$ 77  | 7461 $\pm$ 113 | 7855 $\pm$ 47 | 6313 $\pm$ 6 | 6363 $\pm$ 6 | 6239 $\pm$ 10 | 6258 $\pm$ 8 | 6396 $\pm$ 11 | 6587 $\pm$ 11 |
|     | 20  | 5402 $\pm$ 21 | 6266 $\pm$ 42 | 6216 $\pm$ 44  | 7407 $\pm$ 56  | 7329 $\pm$ 108 | 7391 $\pm$ 30 | 6313 $\pm$ 6 | 6363 $\pm$ 6 | 6239 $\pm$ 10 | 6258 $\pm$ 8 | 6396 $\pm$ 11 | 6587 $\pm$ 11 |
| 105 | 5   | 3970 $\pm$ 33 | 6535 $\pm$ 56 | 6350 $\pm$ 73  | 8276 $\pm$ 126 | 8292 $\pm$ 89  | 8340 $\pm$ 35 | 6809 $\pm$ 5 | 6850 $\pm$ 4 | 6783 $\pm$ 4  | 6793 $\pm$ 6 | 6743 $\pm$ 13 | 6874 $\pm$ 12 |
|     | 10  | 4951 $\pm$ 37 | 6560 $\pm$ 43 | 6297 $\pm$ 55  | 7801 $\pm$ 173 | 7927 $\pm$ 154 | 8200 $\pm$ 61 | 6809 $\pm$ 5 | 6850 $\pm$ 4 | 6783 $\pm$ 4  | 6793 $\pm$ 6 | 6743 $\pm$ 13 | 6874 $\pm$ 12 |
|     | 20  | 5697 $\pm$ 23 | 6636 $\pm$ 32 | 6566 $\pm$ 40  | 7562 $\pm$ 103 | 7688 $\pm$ 90  | 7674 $\pm$ 27 | 6809 $\pm$ 5 | 6850 $\pm$ 4 | 6783 $\pm$ 4  | 6793 $\pm$ 6 | 6743 $\pm$ 13 | 6874 $\pm$ 12 |
| 200 | 5   | 3939 $\pm$ 27 | 6787 $\pm$ 52 | 6516 $\pm$ 101 | 8365 $\pm$ 102 | 8450 $\pm$ 118 | 8420 $\pm$ 46 | 7396 $\pm$ 6 | 7421 $\pm$ 6 | 7404 $\pm$ 4  | 7417 $\pm$ 7 | 7134 $\pm$ 12 | 7203 $\pm$ 7  |
|     | 10  | 4998 $\pm$ 48 | 6844 $\pm$ 57 | 6576 $\pm$ 36  | 8383 $\pm$ 131 | 8163 $\pm$ 135 | 8513 $\pm$ 50 | 7396 $\pm$ 6 | 7421 $\pm$ 6 | 7404 $\pm$ 4  | 7417 $\pm$ 7 | 7134 $\pm$ 12 | 7203 $\pm$ 7  |
|     | 20  | 5871 $\pm$ 25 | 6995 $\pm$ 47 | 6844 $\pm$ 53  | 7296 $\pm$ 118 | 7605 $\pm$ 117 | 7627 $\pm$ 58 | 7396 $\pm$ 6 | 7421 $\pm$ 6 | 7404 $\pm$ 4  | 7417 $\pm$ 7 | 7134 $\pm$ 12 | 7203 $\pm$ 7  |

Table 14: **Warfarin** ( $d=93$ ,  $K=8$ ,  $T=5,000$ ) per-rank results across  $r \in \{1, 2, 3, 5, 10\}$ . 10 seeds.

| $d$ | $r$ | Oracle         | SPSC          | SPSC-Adp      | LowOFUL        | VOFUL         | LR-Rew        | SW-Lin        | D-Lin         | Rst-Lin       | LinUCB        | LinTS         | SW-LinTS      |
|-----|-----|----------------|---------------|---------------|----------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| 93  | 1   | 6.4 $\pm$ 0.7  | 1482 $\pm$ 49 | 626 $\pm$ 44  | 1491 $\pm$ 26  | 1497 $\pm$ 23 | 1533 $\pm$ 20 | 2096 $\pm$ 18 | 2175 $\pm$ 22 | 1977 $\pm$ 15 | 2006 $\pm$ 17 | 1949 $\pm$ 20 | 2060 $\pm$ 22 |
|     | 2   | 93.9 $\pm$ 1.3 | 1404 $\pm$ 35 | 706 $\pm$ 37  | 1076 $\pm$ 130 | 918 $\pm$ 127 | 1140 $\pm$ 25 | 2096 $\pm$ 18 | 2175 $\pm$ 22 | 1977 $\pm$ 15 | 2006 $\pm$ 17 | 1949 $\pm$ 20 | 2060 $\pm$ 22 |
|     | 3   | 177 $\pm$ 4    | 1372 $\pm$ 33 | 657 $\pm$ 39  | 683 $\pm$ 97   | 779 $\pm$ 93  | 1056 $\pm$ 40 | 2096 $\pm$ 18 | 2175 $\pm$ 22 | 1977 $\pm$ 15 | 2006 $\pm$ 17 | 1949 $\pm$ 20 | 2060 $\pm$ 22 |
|     | 5   | 272 $\pm$ 7    | 1369 $\pm$ 30 | 769 $\pm$ 37  | 905 $\pm$ 110  | 930 $\pm$ 76  | 1178 $\pm$ 32 | 2096 $\pm$ 18 | 2175 $\pm$ 22 | 1977 $\pm$ 15 | 2006 $\pm$ 17 | 1949 $\pm$ 20 | 2060 $\pm$ 22 |
|     | 10  | 569 $\pm$ 4    | 1481 $\pm$ 19 | 1014 $\pm$ 35 | 1441 $\pm$ 63  | 1680 $\pm$ 59 | 2078 $\pm$ 19 | 2096 $\pm$ 18 | 2175 $\pm$ 22 | 1977 $\pm$ 15 | 2006 $\pm$ 17 | 1949 $\pm$ 20 | 2060 $\pm$ 22 |

Table 15: **Open Bandit (ZOZOTOWN)** per-cell results.  $T=5,000$ ,  $K=10$ , 10 seeds. Mean  $\pm$  SE costed regret. Green = SPSC variant beats every non-oracle baseline; underline = best non-oracle. LinTS / SW-LinTS not run on this benchmark.

| $d$ | $r$ | Oracle         | SPSC            | SPSC-Adp        | LowOFUL         | VOFUL           | LR-Rew          | SW-Lin          | D-Lin           | Rst-Lin         | LinUCB          |
|-----|-----|----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| 55  | 5   | 34.2 $\pm$ 0.5 | 83.0 $\pm$ 2.3  | 91.1 $\pm$ 4.0  | 126.2 $\pm$ 1.8 | 126.7 $\pm$ 1.5 | 126.8 $\pm$ 1.2 | 103.0 $\pm$ 0.5 | 106.6 $\pm$ 0.5 | 101.9 $\pm$ 0.6 | 102.2 $\pm$ 0.5 |
|     | 10  | 65.2 $\pm$ 0.8 | 93.9 $\pm$ 1.5  | 94.1 $\pm$ 1.7  | 129.3 $\pm$ 2.6 | 130.2 $\pm$ 1.9 | 130.9 $\pm$ 1.3 | 103.0 $\pm$ 0.5 | 106.6 $\pm$ 0.5 | 101.9 $\pm$ 0.6 | 102.2 $\pm$ 0.5 |
| 105 | 5   | 29.7 $\pm$ 0.5 | 106.6 $\pm$ 4.8 | 103.3 $\pm$ 2.9 | 152.6 $\pm$ 2.3 | 150.1 $\pm$ 1.9 | 150.9 $\pm$ 1.6 | 107.4 $\pm$ 1.1 | 108.4 $\pm$ 0.4 | 106.4 $\pm$ 0.5 | 107.7 $\pm$ 1.0 |
|     | 10  | 59.0 $\pm$ 0.7 | 106.9 $\pm$ 3.0 | 106.3 $\pm$ 3.2 | 153.3 $\pm$ 1.0 | 155.4 $\pm$ 1.8 | 156.1 $\pm$ 1.5 | 107.4 $\pm$ 1.1 | 108.4 $\pm$ 0.4 | 106.4 $\pm$ 0.5 | 107.7 $\pm$ 1.0 |

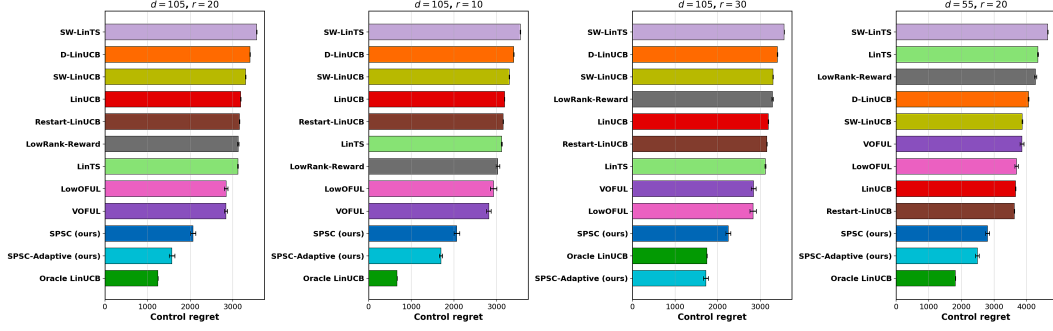


Figure 2: **Pendigits operating-regime grid.** SPSC and SPSC-Adaptive dominate every non-oracle baseline once  $d \geq 55$  and  $r \geq 10$ .

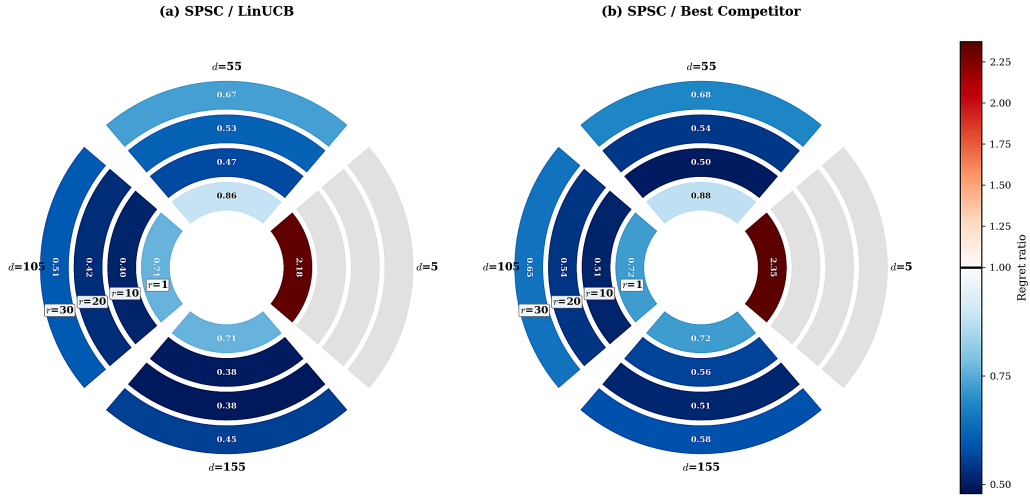


Figure 3: **Satimage operating-regime sunburst.** Regret ratios across the  $(d, r)$  grid. Blue cells (ratio  $< 1$ ) indicate SPSC wins; red cells indicate losses. (a) SPSC vs. LinUCB; (b) SPSC vs. best non-oracle competitor.

830 **Reading the tables.** Three qualitative patterns are consistent across datasets. (i) *Rank-*  
831 *insensitive competitors.* LinUCB, SW-LinUCB, D-LinUCB and Restart-LinUCB operate in  
832 ambient space and therefore produce the *same* regret in every row of a  $d$  block. When a  
833 row shows all four of these at identical values, this is by design: it reflects the  $r$ -insensitivity  
834 of ambient-rate baselines. (ii) *Subspace-aware competitors* (LowOFUL, VOFUL, LowRank-  
835 Reward) do improve with  $d$  increasing but they assume a fixed subspace and therefore  
836 still trail SPSC on segments where the subspace changes. (iii) *Within the synthetic and*  
837 *UCI grids, SPSC-Adaptive typically trails SPSC-Alg. 1 by a small margin;* this gap reflects  
838 the detector’s false-alarm budget. The ordering reverses on the clinical benchmarks (most  
839 strikingly Vancomycin at  $r=1$ , where SPSC-Adaptive cuts regret by 37% vs. SPSC-Alg. 1;  
840 see Table 19), where the oracle schedule of Alg. 1 is misaligned with the true cohort change  
841 points and the detector’s adaptivity wins.

842 **Synthetic  $(d, r)$  grid.** Table 16 gives the full synthetic grid underlying Figure 1. SPSC  
843 wins on 37/40 cells; losses are confined to  $d \leq 10$  or  $r \leq 1$  (probe-cost dominant).

Table 16: **Synthetic grid (full 40 cells).** Mean costed regret across 10 seeds for 10 methods over  $(d, r) \in \{5, 10, 20, 30, 45, 60, 80, 100\} \times \{1, 3, 5, 10, 15, 20\}$  with  $r < d$ ,  $T=5,000$ ,  $K=10$ , probe period 50, window  $W=400$ ,  $\lambda=0.01$ . **Methods:** SPSC (Algorithm 1, ours); **Adpt** = SPSC-Adaptive (Algorithm 2, ours); LinUCB; **Oracle** = Oracle-LinUCB (given the true subspace); **SW-Lin** = SW-LinUCB; **D-Lin** = D-LinUCB; **Rst** = Restart-LinUCB; **LR-Rew** = LowRank-Reward; LowOFUL; VOFUL. S/L = SPSC/LinUCB; S/Best = SPSC/ $\min\{\text{non-oracle, non-Adpt baselines}\}$ . **Verdict.** “SPSC” if  $S/L < 1$  (SPSC has lower regret than LinUCB); “LinUCB” otherwise. Bold = SPSC verdict. Counts: **37 SPSC**, **3 LinUCB**; the 3 LinUCB cells are all in the small- $d$  corner  $d \leq 10$  with  $r$  close to  $d$ , where the probe cost dominates the dimensionality savings.

| $d$ | $r$ | SPSC       | Adpt | LinUCB | Oracle | SW-Lin | D-Lin | Rst  | LR-Rew | LowOFUL | VOFUL | S/L          | S/Best       | verdict     |
|-----|-----|------------|------|--------|--------|--------|-------|------|--------|---------|-------|--------------|--------------|-------------|
| 5   | 1   | 355        | 344  | 203    | 8      | 213    | 234   | 206  | 416    | 514     | 553   | 1.742        | 1.742        | LinUCB      |
| 5   | 3   | 339        | 384  | 221    | 134    | 221    | 254   | 217  | 629    | 930     | 889   | 1.529        | 1.563        | LinUCB      |
| 5   | 10  | 270        | 289  | 255    | 24     | 260    | 271   | 258  | 254    | 317     | 315   | 1.055        | 1.060        | LinUCB      |
| 10  | 3   | <b>389</b> | 422  | 399    | 130    | 391    | 428   | 386  | 456    | 629     | 595   | <b>0.976</b> | 1.008        | <b>SPSC</b> |
| 10  | 5   | <b>465</b> | 525  | 487    | 254    | 482    | 523   | 471  | 660    | 800     | 823   | <b>0.955</b> | 0.988        | <b>SPSC</b> |
| 20  | 1   | <b>231</b> | 228  | 251    | 21     | 251    | 253   | 248  | 179    | 238     | 240   | <b>0.923</b> | 1.290        | <b>SPSC</b> |
| 20  | 3   | <b>353</b> | 377  | 456    | 134    | 449    | 467   | 451  | 344    | 449     | 475   | <b>0.774</b> | 1.026        | <b>SPSC</b> |
| 20  | 5   | <b>420</b> | 453  | 565    | 232    | 555    | 581   | 558  | 468    | 575     | 578   | <b>0.744</b> | <b>0.898</b> | <b>SPSC</b> |
| 20  | 10  | <b>631</b> | 699  | 800    | 498    | 780    | 840   | 784  | 804    | 946     | 975   | <b>0.789</b> | <b>0.809</b> | <b>SPSC</b> |
| 20  | 15  | <b>805</b> | 901  | 899    | 736    | 893    | 954   | 883  | 1011   | 1235    | 1222  | <b>0.895</b> | <b>0.912</b> | <b>SPSC</b> |
| 30  | 1   | <b>189</b> | 206  | 220    | 24     | 220    | 222   | 219  | 154    | 198     | 203   | <b>0.859</b> | 1.225        | <b>SPSC</b> |
| 30  | 3   | <b>307</b> | 330  | 407    | 125    | 409    | 413   | 404  | 296    | 355     | 355   | <b>0.754</b> | 1.038        | <b>SPSC</b> |
| 30  | 5   | <b>372</b> | 421  | 544    | 217    | 541    | 553   | 536  | 411    | 498     | 492   | <b>0.683</b> | <b>0.904</b> | <b>SPSC</b> |
| 30  | 10  | <b>572</b> | 615  | 775    | 451    | 763    | 790   | 760  | 690    | 855     | 828   | <b>0.738</b> | <b>0.829</b> | <b>SPSC</b> |
| 30  | 15  | <b>741</b> | 810  | 930    | 663    | 917    | 957   | 913  | 956    | 1066    | 1113  | <b>0.796</b> | <b>0.811</b> | <b>SPSC</b> |
| 30  | 20  | <b>890</b> | 960  | 1018   | 838    | 1002   | 1056  | 1009 | 1109   | 1236    | 1259  | <b>0.874</b> | <b>0.888</b> | <b>SPSC</b> |
| 45  | 1   | <b>165</b> | 169  | 188    | 25     | 187    | 189   | 186  | 130    | 172     | 168   | <b>0.881</b> | 1.275        | <b>SPSC</b> |
| 45  | 3   | <b>271</b> | 296  | 371    | 118    | 371    | 375   | 369  | 273    | 312     | 294   | <b>0.730</b> | <b>0.992</b> | <b>SPSC</b> |
| 45  | 5   | <b>360</b> | 389  | 504    | 212    | 506    | 510   | 502  | 376    | 419     | 431   | <b>0.715</b> | <b>0.957</b> | <b>SPSC</b> |
| 45  | 10  | <b>499</b> | 520  | 675    | 395    | 665    | 684   | 668  | 559    | 646     | 650   | <b>0.739</b> | <b>0.893</b> | <b>SPSC</b> |
| 45  | 15  | <b>651</b> | 684  | 819    | 577    | 812    | 836   | 811  | 736    | 811     | 808   | <b>0.796</b> | <b>0.885</b> | <b>SPSC</b> |
| 45  | 20  | <b>811</b> | 836  | 973    | 758    | 954    | 980   | 952  | 901    | 988     | 989   | <b>0.833</b> | <b>0.900</b> | <b>SPSC</b> |
| 60  | 1   | <b>141</b> | 146  | 165    | 30     | 166    | 166   | 165  | 108    | 134     | 134   | <b>0.858</b> | 1.311        | <b>SPSC</b> |
| 60  | 3   | <b>229</b> | 243  | 296    | 117    | 294    | 296   | 292  | 222    | 249     | 239   | <b>0.776</b> | 1.033        | <b>SPSC</b> |
| 60  | 5   | <b>315</b> | 319  | 414    | 199    | 412    | 416   | 414  | 320    | 366     | 343   | <b>0.760</b> | <b>0.985</b> | <b>SPSC</b> |
| 60  | 10  | <b>441</b> | 479  | 605    | 367    | 600    | 608   | 601  | 536    | 564     | 585   | <b>0.729</b> | <b>0.822</b> | <b>SPSC</b> |
| 60  | 15  | <b>588</b> | 605  | 743    | 524    | 740    | 752   | 738  | 713    | 752     | 736   | <b>0.791</b> | <b>0.824</b> | <b>SPSC</b> |
| 60  | 20  | <b>708</b> | 753  | 853    | 663    | 845    | 868   | 845  | 838    | 882     | 862   | <b>0.830</b> | <b>0.845</b> | <b>SPSC</b> |
| 80  | 1   | <b>110</b> | 125  | 118    | 27     | 118    | 118   | 118  | 91     | 111     | 106   | <b>0.931</b> | 1.214        | <b>SPSC</b> |
| 80  | 3   | <b>231</b> | 250  | 290    | 116    | 292    | 293   | 290  | 224    | 238     | 239   | <b>0.796</b> | 1.034        | <b>SPSC</b> |
| 80  | 5   | <b>279</b> | 295  | 381    | 187    | 379    | 384   | 379  | 293    | 307     | 319   | <b>0.733</b> | <b>0.954</b> | <b>SPSC</b> |
| 80  | 10  | <b>419</b> | 430  | 542    | 345    | 539    | 545   | 538  | 447    | 493     | 473   | <b>0.772</b> | <b>0.937</b> | <b>SPSC</b> |
| 80  | 15  | <b>534</b> | 576  | 671    | 486    | 668    | 671   | 667  | 587    | 638     | 655   | <b>0.796</b> | <b>0.911</b> | <b>SPSC</b> |
| 80  | 20  | <b>640</b> | 663  | 767    | 599    | 761    | 770   | 761  | 691    | 744     | 761   | <b>0.835</b> | <b>0.927</b> | <b>SPSC</b> |
| 100 | 1   | <b>107</b> | 111  | 112    | 28     | 113    | 113   | 113  | 79     | 86      | 93    | <b>0.954</b> | 1.359        | <b>SPSC</b> |
| 100 | 3   | <b>192</b> | 194  | 238    | 107    | 236    | 238   | 236  | 176    | 193     | 201   | <b>0.809</b> | 1.094        | <b>SPSC</b> |
| 100 | 5   | <b>272</b> | 273  | 346    | 183    | 343    | 346   | 344  | 266    | 294     | 290   | <b>0.787</b> | 1.024        | <b>SPSC</b> |
| 100 | 10  | <b>368</b> | 388  | 471    | 317    | 472    | 472   | 470  | 403    | 438     | 440   | <b>0.780</b> | <b>0.912</b> | <b>SPSC</b> |
| 100 | 15  | <b>466</b> | 491  | 573    | 430    | 571    | 576   | 572  | 517    | 530     | 535   | <b>0.813</b> | <b>0.902</b> | <b>SPSC</b> |
| 100 | 20  | <b>609</b> | 634  | 726    | 572    | 725    | 733   | 723  | 661    | 692     | 702   | <b>0.839</b> | <b>0.921</b> | <b>SPSC</b> |

## 844 G Sensitivity and robustness studies

### 845 G.1 Probe-rate ablation

846 Sweeping the probe period over  $\{5, 10, 20, 30, 50, 100, 300\}$  at  $d=4$ ,  $r=1$ ,  $K=4$ ,  $T=6,000$   
 847 produces a clearly U-shaped regret curve (Figure 4): sparse probing under-recovers the  
 848 subspace; over-frequent probing sacrifices exploitation. Optimum at probe period 20–30,  
 849 matching  $m_k^* \propto \ell_k^{2/3}$  prediction. The right panel shows a monotone relationship between  
 850 late-segment subspace error and final regret, directly linking SPSC’s gains to subspace  
 851 recovery quality.

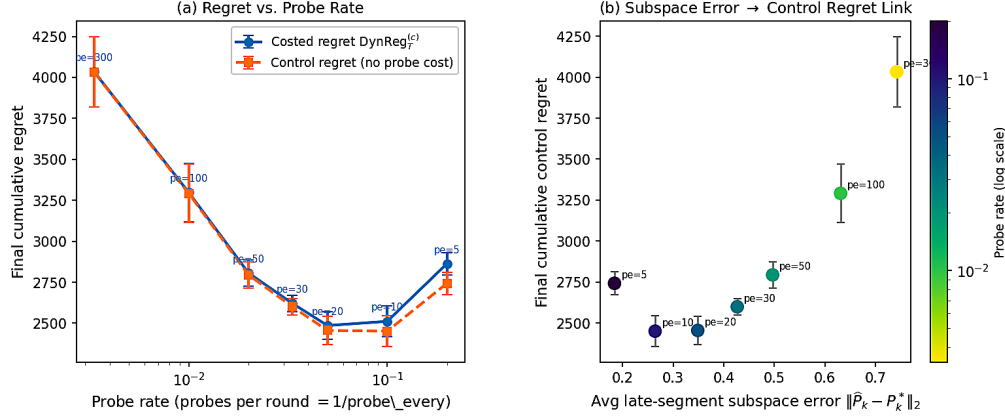


Figure 4: Probe-rate ablation. Left: final regret vs. probe frequency. Right: final control regret vs. late-segment subspace error.

### 852 G.2 Rank misspecification

853 On Covertypes ( $d=155$ , true effective rank  $r^*=10$ ,  $K=4$ ,  $T=10,000$ , 5 seeds), sweeping spec-  
 854 ified  $r$  at the grid points  $\{1, 3, 5, 10, 15, 20, 30, 50, 80\}$  (min  $r=1$ , max  $r=80$ ) gives the U-  
 855 shaped curve in Figure 5. Underestimation loses signal (+33% at  $r=1$ ); mild overestima-  
 856 tion captures residual structure (sweet spot at  $r=30$ , −10%). SPSC beats LinUCB across  
 857  $r \in [15, 50]$ , so mild overestimation is a safe default.

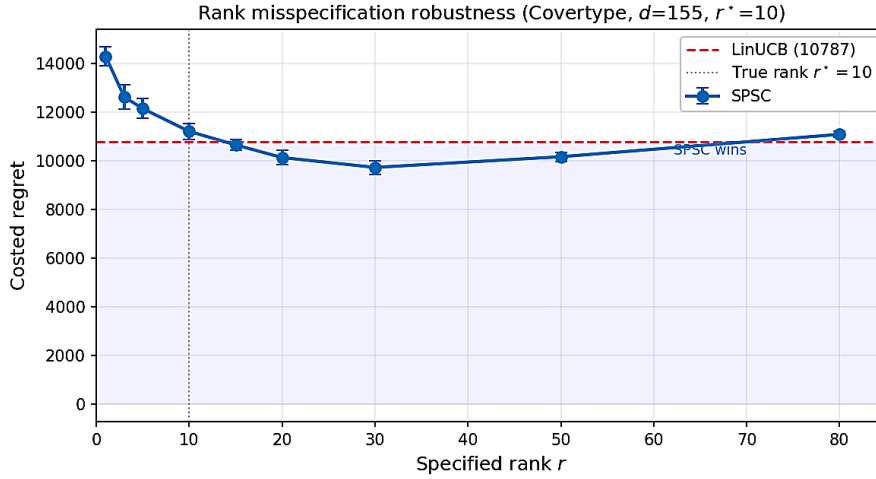


Figure 5: Rank misspecification robustness (Covertypes,  $d=155$ ,  $r^*=10$ ). SPSC beats LinUCB for  $r \in [15, 50]$ ; sweet spot at  $r=30$  (10% improvement).

### G.3 Robustness and necessity experiments (A/B/C)

Three ablations on the reference setting ( $d=4$ ,  $r=1$ ,  $K=4$ ,  $T=6,000$ , probe period 30,  $W=100$ , 10 seeds).

**A. Variance misspecification.** Injecting  $|\delta_\sigma| \in \{0, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5\}$  into the probe statistic  $s_t = y_t^2 - (\sigma_\varepsilon^2 + \delta_\sigma)$ . Regret and subspace error are stable through  $|\delta_\sigma| = 0.5$  (more than  $5\times$  the true variance  $\sigma_\varepsilon^2 = 0.09$ ); at maximum misspecification, regret changes by  $< 5\%$ . For scaled-sphere probes (the theory probe distribution), a constant centering error contributes the population-level identity shift  $\tilde{B} = -(\delta_\sigma/d)I_d$ , which uniformly shifts eigenvalues without corrupting eigenvectors.

**B. Bounded cross-correlation.** Injecting  $\varepsilon_t \rightarrow \varepsilon_t + \epsilon_\times x_t^\top \theta_t$  for  $\epsilon_\times \in \{0, 0.01, 0.05, 0.1, 0.2, 0.5, 1.0\}$ . Even at  $\epsilon_\times = 1.0$  (cross-correlation equal to signal), regret increases by  $< 1\%$ , indicating that moderate violations of orthogonality are practically benign.

**C. Imperfect probe coverage.** Restricting probe directions to the first  $d_{\text{cov}} \in \{1, 2, 3, 4\}$  coordinates produces a dramatic monotonic regret blow-up: at  $d_{\text{cov}} = 1$ , SPSC regret rises to 4,289, nearly matching LinUCB (4,463); subspace error degrades to 0.75. This directly confirms the necessity results: restricted coverage destroys identifiability and forces  $\Omega(T)$  regret within the unobserved directions.

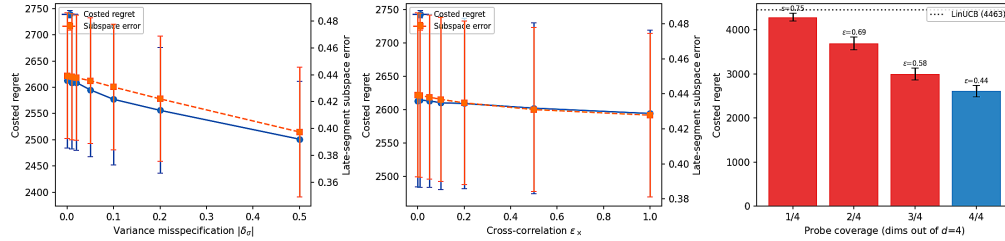


Figure 6: Robustness and necessity experiments. (a) Variance misspecification, regret stable through  $|\delta_\sigma| = 0.5$ . (b) Cross-correlation relaxation,  $< 1\%$  degradation through  $\epsilon_\times = 1.0$ . (c) Imperfect coverage, regret blow-up with restricted probe directions.

### G.4 Assumption-violation master table (large-scale)

We complement the small-scale A/B/C ablations above with a large-scale ( $d=60$ ,  $r=5$ ,  $K=10$ ,  $T=5,000$ ) assumption sweep over (a) variance misspecification  $\hat{\sigma}^2/\sigma_\varepsilon^2 \in \{0.5, 1, 2, 4\}$ , (b) approximate-rank perturbation  $\epsilon_k^\perp \in \{0, 0.05, 0.1, 0.2, 0.4\}$ , and (c) spectral radius  $\rho(A_k) \in \{0.8, 0.95, 0.99, 1.0\}$  (the last violates the stable-LDS hypothesis). Variance misspec is absorbed into the scaled-identity bias (Lemma B.6) and the relative gap stays flat ( $\hat{\sigma}^2/\sigma_\varepsilon^2$  ratio  $1.27 \rightarrow 1.16$  as the sweep goes  $0.5 \rightarrow 4$ ); approximate-rank degrades regret near-linearly in  $\epsilon_k^\perp$ ; SPSC remains well-behaved through  $\rho = 0.99$  and degrades gracefully at  $\rho = 1$  (random-walk limit, outside the stable-LDS assumption).

### G.5 Subspace recovery rate and change-point adaptation

Figure 7 plots the binned mean subspace error  $\|\hat{P}_k - P_k^*\|_2$  vs. probe count; the empirical rate matches the predicted  $1/\sqrt{m}$  on a log-log scale. Figure 8 plots post-change instantaneous regret and the subspace-error trajectory: SPSC recovers substantially faster than ambient LinUCB after each boundary.

### G.6 Noise robustness, change-point frequency, drift speed

Three single-parameter sweeps on the reference setting.

Table 17: **Assumption-violation master table** ( $d=60, r=5, K=10, T=5,000$ , 10 seeds, costed dynamic regret, mean $\pm$ SE). SPSC tracks Oracle through all three sweeps; bold = best non-Oracle in row.

| Sweep                              | Level | Oracle        | SPSC-Alg1                       | SPSC-Adap                       | LinUCB        | SW-LinUCB     | D-LinUCB      | LowOFUL                        |
|------------------------------------|-------|---------------|---------------------------------|---------------------------------|---------------|---------------|---------------|--------------------------------|
| $\hat{\sigma}^2/\sigma_\epsilon^2$ | 0.5   | 221 $\pm$ 5   | <b>281 <math>\pm</math> 10</b>  | 304 $\pm$ 11                    | 397 $\pm$ 17  | 395 $\pm$ 16  | 401 $\pm$ 18  | 370 $\pm$ 19                   |
|                                    | 1.0   | 257 $\pm$ 6   | <b>315 <math>\pm</math> 10</b>  | 319 $\pm$ 10                    | 413 $\pm$ 18  | 412 $\pm$ 18  | 415 $\pm$ 19  | 358 $\pm$ 17                   |
|                                    | 2.0   | 299 $\pm$ 10  | 357 $\pm$ 13                    | <b>355 <math>\pm</math> 13</b>  | 421 $\pm$ 19  | 421 $\pm$ 18  | 423 $\pm$ 20  | 377 $\pm$ 14                   |
|                                    | 4.0   | 344 $\pm$ 12  | 398 $\pm$ 16                    | 398 $\pm$ 16                    | 426 $\pm$ 19  | 425 $\pm$ 19  | 427 $\pm$ 20  | <b>390 <math>\pm</math> 20</b> |
| $\epsilon_k^\perp$                 | 0.00  | 257 $\pm$ 6   | <b>315 <math>\pm</math> 10</b>  | 319 $\pm$ 10                    | 413 $\pm$ 18  | 412 $\pm$ 18  | 415 $\pm$ 19  | 358 $\pm$ 17                   |
|                                    | 0.05  | 255 $\pm$ 7   | <b>311 <math>\pm</math> 12</b>  | 331 $\pm$ 10                    | 414 $\pm$ 18  | 414 $\pm$ 18  | 417 $\pm$ 19  | 379 $\pm$ 20                   |
|                                    | 0.10  | 261 $\pm$ 7   | <b>319 <math>\pm</math> 13</b>  | 331 $\pm$ 12                    | 422 $\pm$ 19  | 422 $\pm$ 18  | 425 $\pm$ 19  | 376 $\pm$ 20                   |
|                                    | 0.20  | 286 $\pm$ 8   | <b>340 <math>\pm</math> 13</b>  | 348 $\pm$ 6                     | 452 $\pm$ 19  | 453 $\pm$ 18  | 457 $\pm$ 20  | 399 $\pm$ 19                   |
|                                    | 0.40  | 358 $\pm$ 9   | <b>396 <math>\pm</math> 10</b>  | 410 $\pm$ 10                    | 559 $\pm$ 21  | 561 $\pm$ 20  | 565 $\pm$ 22  | 497 $\pm$ 13                   |
| $\rho(A_k)$                        | 0.80  | 713 $\pm$ 6   | 745 $\pm$ 7                     | 747 $\pm$ 9                     | 744 $\pm$ 8   | 743 $\pm$ 6   | 745 $\pm$ 8   | <b>732 <math>\pm</math> 9</b>  |
|                                    | 0.95  | 1083 $\pm$ 11 | <b>1291 <math>\pm</math> 16</b> | 1307 $\pm$ 14                   | 1403 $\pm$ 16 | 1406 $\pm$ 15 | 1404 $\pm$ 19 | 1330 $\pm$ 18                  |
|                                    | 0.99  | 1132 $\pm$ 23 | 2017 $\pm$ 46                   | <b>1929 <math>\pm</math> 34</b> | 2751 $\pm$ 46 | 2732 $\pm$ 44 | 2750 $\pm$ 52 | 2585 $\pm$ 54                  |
|                                    | 1.00  | 914 $\pm$ 19  | 2899 $\pm$ 83                   | <b>2813 <math>\pm</math> 59</b> | 4644 $\pm$ 90 | 4523 $\pm$ 87 | 4674 $\pm$ 90 | 5552 $\pm$ 156                 |

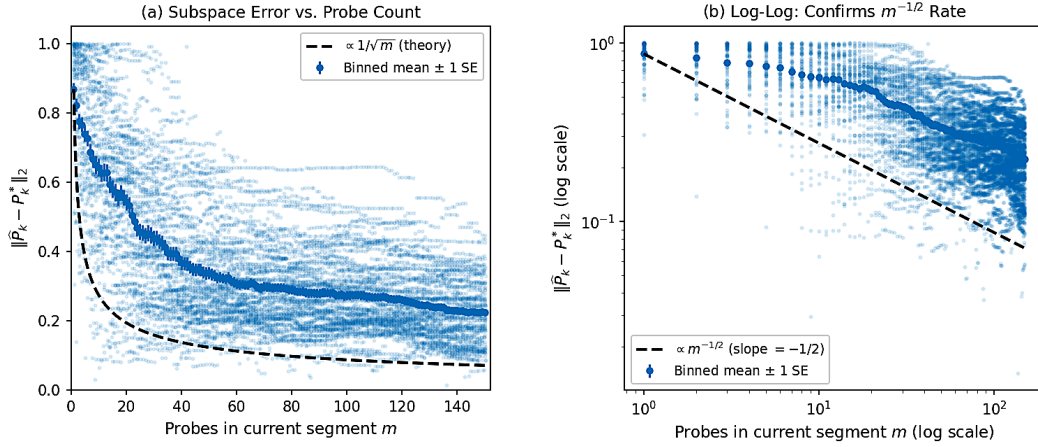


Figure 7: **Subspace recovery.**  $\|\hat{P}_k - P_k^*\|_2$  vs. probe count matches the predicted  $1/\sqrt{m}$  rate on a log-log scale.

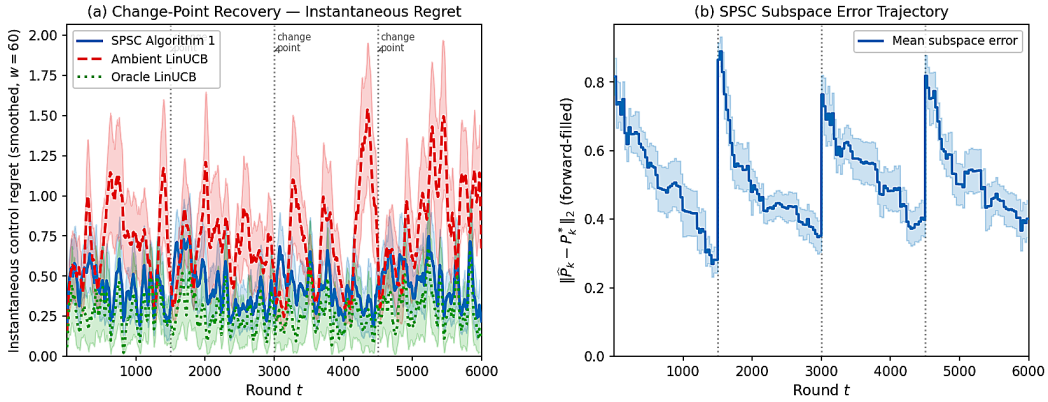


Figure 8: **Change-point adaptation.** Instantaneous regret drops sharply after each boundary as fresh probes rebuild the subspace.

**Noise.** Sweeping  $\sigma_\varepsilon \in \{0.05, 0.10, 0.20, 0.30, 0.50, 0.80\}$ : SPSC/LinUCB ratio grows mildly with noise (from 0.566 to 0.595); SPSC/Oracle stable at  $\approx 1.55\times$ . The low-rank benefit is not a low-noise artifact.

**Change-point frequency.**  $K \in \{1, 2, 4, 6, 8, 12\}$  at  $T=6,000$ . Ratio grows from 0.414 (long segments) to 0.855 (very short); SPSC retains an edge at all frequencies but the margin shrinks as the per-segment probe budget collapses below the identifiability threshold.

**Drift speed.** LDS spectral radius  $\rho \in \{0.30, \dots, 0.99\}$ . Sharp phase transition: SPSC/LinUCB  $\approx 1.01$  for  $\rho \leq 0.70$ , drops to 0.861 at  $\rho=0.95$ , and to 0.582 at  $\rho=0.99$ . Subspace error is nearly constant across  $\rho$  (0.28–0.33), so the missing benefit at low  $\rho$  is a signal-energy effect, not estimation failure: low-rank structure in a signal already dominated by noise leaves little to extract.

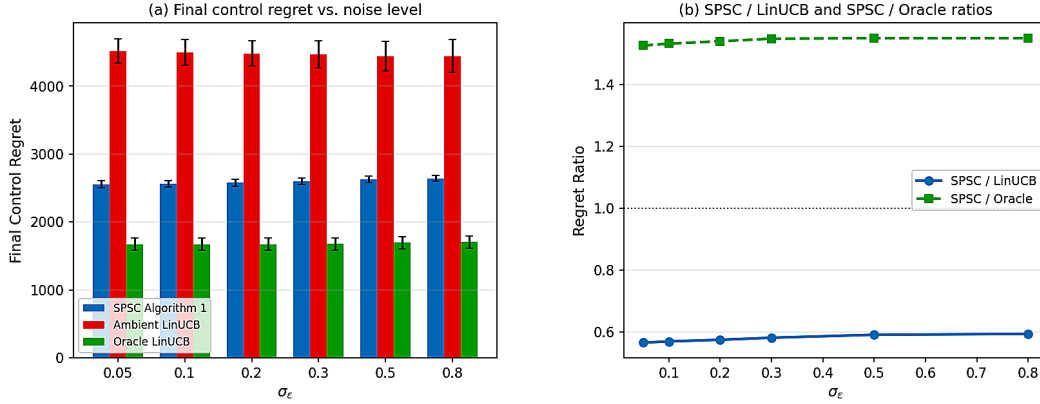


Figure 9: Noise robustness: SPSC advantage is noise-monotone.

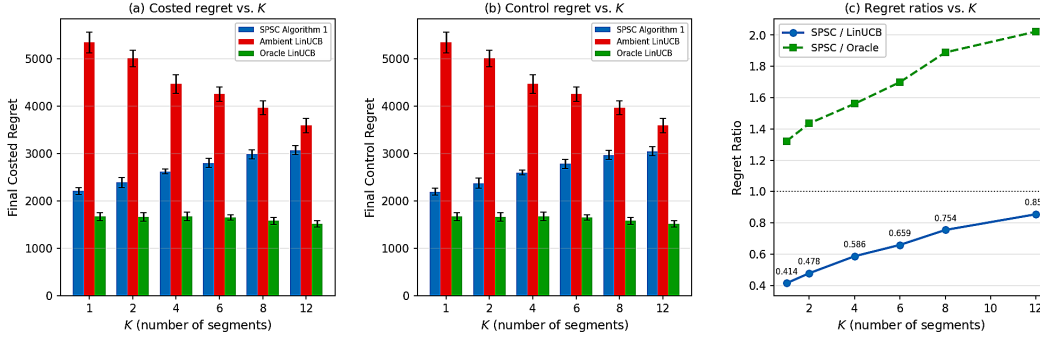


Figure 10: Change-point frequency.

## H Comparison with nonstationary baselines (Russac benchmark)

**Russac et al. NeurIPS 2019 benchmark.**  $d=2, r=1, K=4, T=6,000, \sigma_\varepsilon=1, 50$  arms, 30 seeds. SPSC reduces control regret by 46.5% vs. D-LinUCB and 34.8% vs. SW-LinUCB; probe overhead is a 0.9% fraction.

## I Warfarin: random-subspace ablation

To decompose SPSC’s Warfarin gain into (i) dimensionality reduction and (ii) the *learned* subspace itself, we replace  $\hat{U}_t$  with a fresh random rank- $r$  projection every segment and run SPSC otherwise unchanged. Results on  $d=93$  across  $r \in \{1, 2, 3, 5, 10\}$  (10 seeds): the

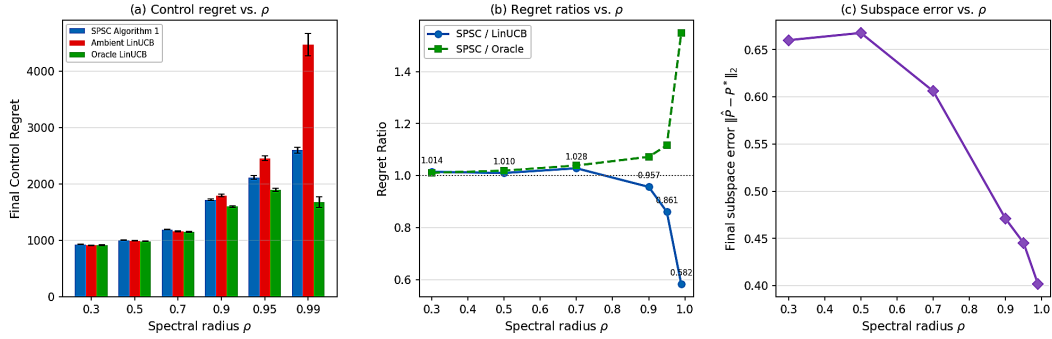


Figure 11: Drift speed. Benefit gated by LDS correlation time  $\tau_\rho = 1/(1 - \rho)$ .

Table 18: Russac et al. benchmark. Final control regret and costed regret (mean $\pm$ SE, 30 seeds).

| Algorithm                          | Control         | Costed | vs. D-LinUCB |
|------------------------------------|-----------------|--------|--------------|
| <b>SPSC Alg. 1 (ours)</b>          | $2,229 \pm 54$  | 2,249  | −46.5%       |
| Oracle LinUCB                      | $1,757 \pm 55$  | 1,757  | −57.8%       |
| D-LinUCB [Russac et al., 2019]     | $4,167 \pm 110$ | 4,167  | —            |
| SW-LinUCB [Cheung et al., 2019]    | $3,417 \pm 82$  | 3,417  | −18.0%       |
| OFUL [Abbasi-Yadkori et al., 2011] | $4,645 \pm 149$ | 4,645  | +11.5%       |
| Reset-LinUCB (oracle)              | $4,647 \pm 147$ | 4,647  | +11.5%       |

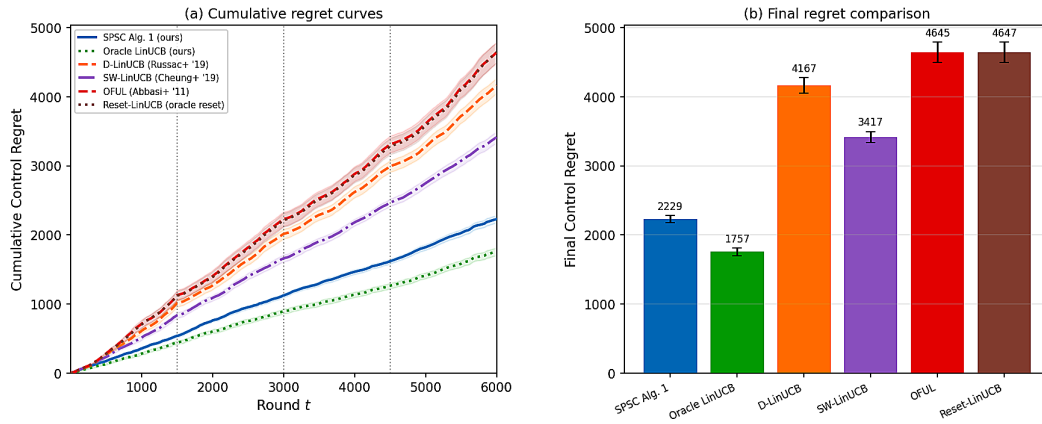


Figure 12: Russac et al. NeurIPS 2019 benchmark.

random-subspace variant retains about half of the SPSC-vs-LinUCB gap, confirming that *dimensionality reduction explains roughly half* the win and the learned subspace explains the rest. The learned subspace captures 2–6 $\times$  more signal variance than a random rank- $r$  projection at every  $r$ , directly quantifying the value of the probe-based identification.

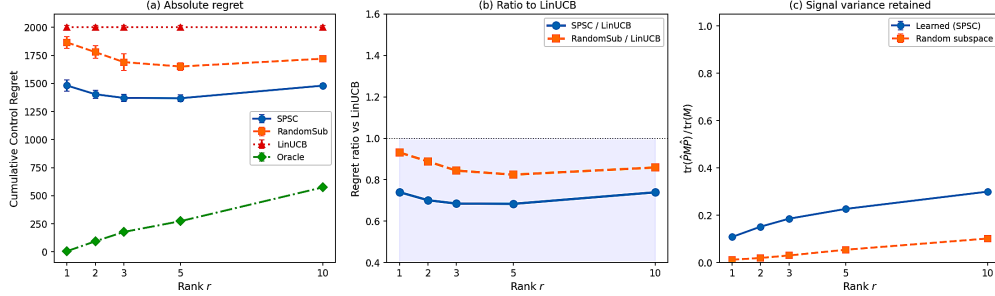


Figure 13: **Warfarin random-subspace ablation.** SPSC’s learned subspace vs. a fresh random rank- $r$  projection at every segment. Half of the gap to LinUCB is dimensionality reduction; the other half is learned signal.

## J BOSS/Jedra adaptation to the piecewise setting

BOSS [Duong et al., 2024] and Jedra [Jedra et al., 2024] are stationary low-rank methods. To give them a fair comparison to SPSC we restart each at the *true* segment boundaries (an oracle advantage we do not give SPSC) and otherwise leave their hyperparameters at the authors’ recommended defaults. The benchmark is piecewise LDS,  $K=10$ ,  $T=5,000$ ,  $\sigma_\varepsilon=0.3$ , 40 actions, 10 seeds, probe cost  $c=0.1$ , on a grid  $d \in \{55, 105, 200\} \times r \in \{5, 10, 20\}$  (eight cells: (55, 5), (55, 10), (105, 5), (105, 10), (105, 20), (200, 5), (200, 10), (200, 20)).

Results (cf. Table 4): SPSC Alg. 1 wins every cell, by 5.8–19.9%, with the lead largest at small  $d$  (where the ambient  $d$  is closest to  $r$  and probing pays off most) and narrowing to 5.8–8.0% at  $d=200$  as oracle-restart-plus-PCA closes the gap. SPSC-Adaptive (no oracle restarts) also beats BOSS and Jedra on every cell, and the ranking is consistent on every seed. The point is that “low-rank exploitation” alone is not enough for changing-subspace problems: the learner must *identify a changing subspace online*, which is SPSC’s core capability.

## K Vancomycin: full per-rank tables

This appendix gives the per-rank Vancomycin results ( $d=93$ ,  $K=8$ ,  $T=5,000$ ,  $\sigma_\varepsilon=0.3$ , 10 seeds, probe period 10, window 200, probe cost  $c=0.1$ ) summarized in Table 3 of the main body. SPSC-Adaptive beats every non-oracle baseline at every rank.

## L When the adaptive variant wins

This appendix disentangles the two SPSC variants: Alg. 1 assumes oracle change points; Alg. 2 detects them online via the detector statistic (14). When the oracle boundaries are correct, Alg. 1 is slightly better because the adaptive detector pays a small statistical price for unknown change-point localization. When the oracle provides misspecified (false-alarm) boundaries, the picture reverses sharply.

SPSC-Adaptive’s regret is flat as  $K_{\text{oracle}}$  grows, while Alg. 1 degrades monotonically and is 3 $\times$  worse at  $K_{\text{oracle}}=50$ : a detector-based reset is strictly safer than a fixed schedule when boundaries are unknown or noisy.

Table 19: **Vancomycin per-rank results** (mean $\pm$ SE, 10 seeds). “vs. best comp.” compares SPSC-Adaptive against the strongest non-oracle baseline in each column (marked \*). SPSC-Adaptive wins every rank.

| Method                       | $r=1$                              | $r=2$                              | $r=3$                              | $r=5$                              | $r=10$                             |
|------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| Oracle LinUCB                | $34.3 \pm 1.5$                     | $115.1 \pm 2.0$                    | $197.2 \pm 3.0$                    | $268.1 \pm 3.4$                    | $373.2 \pm 5.7$                    |
| SPSC Alg. 1 (ours)           | $840.3 \pm 24.5$                   | $805.9 \pm 15.4$                   | $802.3 \pm 12.3$                   | $805.3 \pm 9.5$                    | $811.0 \pm 12.2$                   |
| <b>SPSC-Adaptive</b>         | <b><math>525.3 \pm 45.8</math></b> | <b><math>613.6 \pm 33.1</math></b> | <b><math>640.3 \pm 33.0</math></b> | <b><math>665.9 \pm 27.2</math></b> | <b><math>705.6 \pm 19.5</math></b> |
| LowOFUL                      | $1049.9 \pm 12.8$                  | $740.1 \pm 36.3^*$                 | $734.5 \pm 37.8$                   | $791.5 \pm 26.0$                   | $801.9 \pm 20.1$                   |
| VOFUL                        | $1063.5 \pm 5.5$                   | $789.7 \pm 24.8$                   | $712.5 \pm 25.5^*$                 | $763.0 \pm 30.6^*$                 | $758.6 \pm 27.9^*$                 |
| LowRank-Reward               | $1175.6 \pm 11.6$                  | $806.8 \pm 11.2$                   | $835.1 \pm 10.7$                   | $898.8 \pm 6.7$                    | $880.5 \pm 5.5$                    |
| SW-LinUCB                    | $875.6 \pm 6.7$                    | $875.6 \pm 6.7$                    | $875.6 \pm 6.7$                    | $875.6 \pm 6.7$                    | $875.6 \pm 6.7$                    |
| D-LinUCB                     | $925.3 \pm 5.8$                    | $925.3 \pm 5.8$                    | $925.3 \pm 5.8$                    | $925.3 \pm 5.8$                    | $925.3 \pm 5.8$                    |
| Restart-LinUCB               | $808.9 \pm 5.3^*$                  | $808.9 \pm 5.3$                    | $808.9 \pm 5.3$                    | $808.9 \pm 5.3$                    | $808.9 \pm 5.3$                    |
| LinUCB                       | $825.3 \pm 6.1$                    | $825.3 \pm 6.1$                    | $825.3 \pm 6.1$                    | $825.3 \pm 6.1$                    | $825.3 \pm 6.1$                    |
| SPSC-Adaptive vs. LinUCB     | -36.4%                             | -25.7%                             | -22.4%                             | -19.3%                             | -14.5%                             |
| SPSC-Adaptive vs. best comp. | -35.1%                             | -17.1%                             | -10.1%                             | -12.7%                             | -7.0%                              |

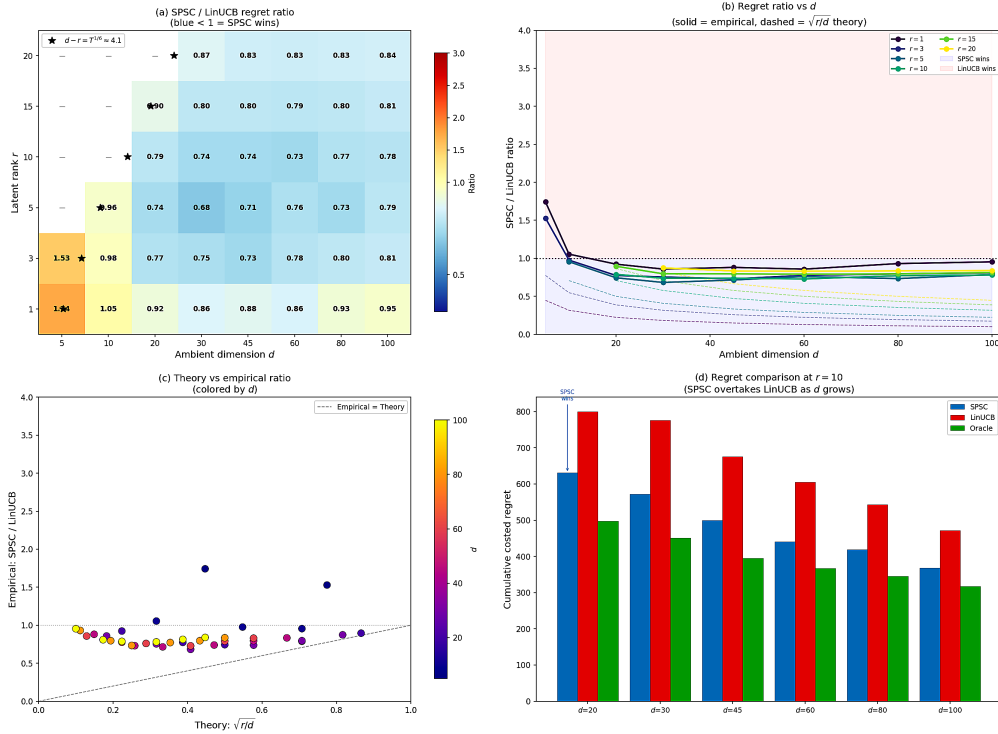


Figure 14: **SPSC-Alg. 1 vs. SPSC-Adaptive**. Top:  $d$ -sweep with correct oracle boundaries. Bottom: oracle quality. With  $K_{\text{real}}=5$  true boundaries, Alg. 1 is fed  $K_{\text{oracle}} \in \{5, \dots, 50\}$  (extras are false alarms) and degrades; Alg. 2 ignores the oracle and is flat.

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

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Justification: The abstract and introduction list four contributions (identification, algorithm, regret bound, experiments) which are stated formally in Theorem 2.2, Algorithms 1–2, Theorem 4.1, and §5; the scope and limitations (e.g. approximate-low-rank settings, oracle vs. detected change points) are flagged in the relevant sections.

### 2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

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Justification: Limitations are stated where they arise: the regret bound is conditional on the windowed projected potential condition (§4); the bound is most informative when within-segment variation  $V_{\text{in}}$  is small (Theorem 4.1 discussion); the rule-of-thumb crossover  $d - r \gtrsim T^{1/6}$  is qualitative, not sharp (§4, §5.1); SPSC-Adaptive is evaluated empirically only with no formal regret guarantee (§4).

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Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

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Justification: All theoretical results (Theorem 2.2, Theorem 4.1, Corollary 4.3, Corollary 4.4, Proposition 4.5) state their assumptions explicitly; full proofs are in Appendix B together with the supporting lemmas (matrix Bernstein, projector concentration, projected nonstationary UCB lemma, prefix concentration).

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Justification: All datasets, dimensions  $d$ , ranks  $r$ , segment counts  $K$ , horizon  $T$ , seed counts, probe rate, regularisation, window length, and per-baseline hyperparameters are listed in §5 and Appendix E; per-cell tables in Appendix F report mean  $\pm$  SE for every dataset,  $(d, r)$  cell, and method.

### 5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

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Justification: The supplemental archive contains all experiment scripts (one per figure and per table), per-dataset environment code, cached JSON results, and a README mapping each figure and table to its generator script.

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Answer: [\[Yes\]](#)

Justification: §5 states the headline parameters per benchmark; Appendix E gives the full hyperparameter table (probe period, ridge  $\lambda$ , window  $W$ , confidence  $\delta$ , probe cost  $c$ , baseline-specific knobs) used uniformly across all  $(d, r)$  cells without per-dataset retuning.

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Justification: Every reported number is  $\text{mean} \pm \text{SE}$  or  $\text{mean} \pm \text{SEM}$  over at least 10 (typically 10–30) seeds; figures show  $\pm 1$  SE bands; the seed count and aggregation choice are stated in each table caption and in §5.

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Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

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Justification: Appendix E reports that the full grid (all benchmarks, 10 seeds, 10 methods) takes 6–20 hours on a single 16-core CPU; no GPU is required (NumPy/SciPy only).

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Justification: The work is a theoretical and empirical study of a stochastic bandit algorithm on synthetic and publicly available benchmarks; the clinical environments are semi-synthetic surrogates calibrated to published cohort statistics, with no patient-identifiable data, no human subjects, and no clinical deployment.

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Justification: The paper studies a foundational sequential-decision algorithm with no specific deployment; the positive impact is improved sample efficiency for low-rank decision problems (recommendation, dosing) and negative-impact pathways are not direct. Clinical bandit benchmarks are evaluation surrogates, not deployable dosing policies.

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 1043 all baseline algorithms (LinUCB, D-LinUCB, SW-LinUCB, LowOFUL, VOFUL,  
 1044 LowRank-Reward, LinTS, BOSS, Jedra) are cited at first use; clinical benchmarks  
 1045 follow the IWPC [Consortium, 2009] and Rybak et al. [Rybak et al., 2020] dosing  
 1046 guidelines.

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1051 Justification: The supplemental material releases the SPSC and SPSC-Adaptive  
 1052 Python implementations, the per-benchmark environments, all experiment scripts,  
 1053 and a README documenting the figure-to-script and table-to-source mappings.

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